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ABSTRACT

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Transforming Cataract Care: Using machine learning to automate triage

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Purpose: Hospitals struggle to efficiently manage patient triage due to growing influx of patients and minimal referral information. This study focuses on leveraging machine learning (ML) to develop an automated triage model for cataract patients, enabling more precise allocation of patients to ophthalmologists and optometrists.
Patients and methods: A comparative analysis was conducted to evaluate the effectiveness of supervised ML algorithms, among which Logistic Regression (LR), in classification of patients to an appointment with an ophthalmologist or optometrist. A "correct" classification was determined by an ophthalmologist appointment with a scheduled/performed surgery afterwards or an optometrist appointment without surgery afterwards within 6 months of the appointment. The dataset consisted primarily of outcomes of a digitally filled in questionnaire, with among others PROMS questions. The supervised ML algorithms were trained and tested on a dataset comprising of 660 patients from June 2023 to June 2024, with validation based on their performance compared to manual triage.
Results: Manual triage has an accuracy of 49% in correctly identifying patients requiring an ophthalmologist or optometrist. The LR algorithm demonstrated superior performance, achieving an accuracy of 68%. Moreover, specificity increased from 28% to 79%, suggesting a more accurate prediction of patient requiring an optometrist appointment.
Conclusions: The findings emphasize the potential of ML-powered triage systems in cataract care in improving accuracy of triage while also automating triage, increasing efficiency of care with minimal involvement of personnel. This study has led to implementation of automated triage, directly after referral, in current patient planning.

Assessing the impact of intraocular lens tilt and decentration on visual acuity

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Purpose: To evaluate the impact of postoperative IOL tilt and decentration on corrected distance visual acuity (CDVA) by performing analyses using IOLMaster700 data and custom-developed software.
Methods: This case series was conducted at the University Eye Clinic Maastricht. Postoperative cataract patients were included and underwent SS-OCT scans using the IOLMaster700 with custom-developed software for tilt and decentration analyses. The outcomes based on our custom-developed software were conducted along the pupillary axis (PA) and corneal topographic axis (CTA) a and the potential impact of tilt and decentration on CDVA was assessed.
Results: The study comprised 168 eyes of 168 patients. The mean magnitude of the tilt along the PA was $1.21 \pm 0.69°$, and decentration was 0.27 ± 0.25 mm. Along the CTA the mean tilt magnitude was $4.87 \pm 1.52°$, and the mean decentration was 0.27 ± 0.26 mm. No statistically significant impact of tilt and decentration on CDVA was observed, with tilt magnitudes up to 3.7° along the PA and 9.5° along the CTA, and mean decentration outcomes of 0.27 mm along both axes.
Conclusion: There was no significant impact of tilt and decentration on CDVA in this cohort. This study underscores the importance of considering specific devices and reference axes when comparing tilt and decentration outcomes, as the methods of analysis cannot be used interchangeably.

Clinical experience with a new refractive extended depth of focus intraocular lens

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Purpose: To evaluate the performance of the new PureSee extended depth of focus (EDF) intraocular lens (IOL).
Methods: So far, 9 patients have been evaluated following cataract surgery with bilateral implantation of the PureSee (ZEN00V, DEN00V, or DETnnn; Johnson & Johnson) IOL in the capsular bag. Binocular uncorrected distance (UDVA), intermediate (UIVA),

and near (UNVA) visual acuity were measured. Reading acuity and speed, assessed in words per minute (wpm), were evaluated using the Radner reading chart. Patient-reported outcomes were gathered using the Catquest-9SF NL, the Patient-Reported Spectacle Independence Questionnaire, and the Quality of Vision questionnaire.
Results: The mean spherical equivalent (SEQ) was +0.08 D in the dominant eye and − 0.01 D in the non-dominant eye. Mean binocular UDVA was −0.06 logMar, UIVA was +0.08 logRAD, and UNVA was +0.28 logRAD. Average reading speeds were 162 wpm at intermediate distance, 163 wpm at near distance, and 184 wpm at near distance with reading glasses. Among the participants, 6/8 (75%) reported being fully spectacle-independent for distance and intermediate vision, while 2/8 (25%) also achieved spectacle independence for near vision. Patients reported experiencing no to minor dysphotopsias.
Conclusions: Early results with the PureSee EDF IOL demonstrate high rates of spectacle independence for distance and intermediate vision, with no to minor reported dysphotopsias reported. However, reading glasses were commonly used for near tasks.

European survey on OCT and anti-VEGF use surrounding cataract surgery

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Purpose: To gain insight into OCT utilization and anti-VEGF use in ophthalmic practice, especially in the context of neovascular Age-related Macular Degeneration (AMD) and cataract surgery.
Methods: An online survey was developed and distributed to 95 ophthalmological clinical and research centres across Europe using the European Vision Clinical Research (EVICR) network. Focusing on ophthalmologists' attitudes towards performing cataract surgery on patients with AMD, their practices regarding anti-VEGF injections in conjunction with cataract surgery, and the integration of OCT in their clinical workflow. The survey aimed to gather insights into the decision-making processes that guide the use of OCT and to identify the level of consensus or variability among different practitioners.
Results: The survey revealed a diverse geographical distribution of respondents (n=72), with a majority practicing in Germany, Italy, and Portugal. Findings revealed that 58.33% of ophthalmologists perform OCT in selected cases before routine cataract surgery, mainly to identify conditions such as epiretinal membrane, vitreomacular traction, and AMD. The most common strategy for managing anti-VEGF injections in relation to cataract surgery was planning surgery in the middle of the interval of two injections. Finally, monofocal IOLs were found to be the preferred choice for patients with neovascular AMD.
Conclusions: We have gained a first overview of OCT use and cataract surgery, and anti-VEGF use and catract surgery. The findings advocate for future research to establish evidence-based practices for OCT utilization and anti-VEGF use.

Dropless cataract surgery: Results of the ESCRS EPICAT study

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Purpose: To evaluate the effectiveness of different dropless strategies to prevent cystoid macular edema (CME) after cataract surgery, using either topical drugs (control group) or intra−/periocular injections (intervention groups).
Methods: The design was a multicentre RCT with a follow up of 12 weeks. Six-hundred-twenty-six patients who required routine cataract surgery were randomized to one of 4 groups: a control group (topical dexamethasone and bromfenac); intervention group 1 (subconjunctival injection of 10 mg triamcinolone acetonide (TA) during cataract surgery); intervention group 2 (intracameral injection of ketorolac solution during cataract surgery); or intervention group 3 (combination of TA and ketorolac). Primary endpoint was the change in central subfield mean macular thickness (CSMT) at 6 weeks postoperatively, as compared to baseline. Secondary endpoints included: incidence of CME; incidence of clinically significant macular edema (CSME); corrected distance visual acuity (CDVA); and intraocular pressure (IOP).
Results: At 6 weeks postoperatively, the CSMT was significantly increased in the ketorolac group compared to the other groups (p < 0.001), whereas the other groups were comparable. The incidence of CME was 2.0% in the control group, 0.7% in the TA and combination groups and 14.0% in the ketorolac group. The incidence of CSME was 5.6% in the ketorolac group compared to 0 or 0.7% in the other groups.
Conclusions: A subconjunctival injection with 10 mg of TA was as effective as topical therapy to decrease macular thickening after cataract surgery and prevent CME. This can be an effective dropless strategy with the potential to save millions of health care costs.

Automated versus subjective/manifest refraction in a novel toric intraocular lens

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Introduction: This study evaluates the correlation between subjective refraction (SR) and automated

refraction (AR) in patients bilaterally implanted with a novel toric pentafocal intraocular lens (IOL).
Patients and Methods: 26 patients (52 eyes) implanted with the Intensity EZ IOL (Hanita Lenses) were included. At the 3-month follow-up, AR and SR were compared in terms of sphere, spherical equivalent (SE), cylinder (C), and astigmatism components J0 and J45. Statistical analysis, including spherical and toric components, was performed using the Eyetemis© tool.
Results: The mean SE difference was +0.68 D±0.33, with AR yielding significantly more myopic values than SR (p<0.001). The sphere difference was +0.52 D±0.32 (p<0.001). The mean difference in cylinder (C) was 0.36±0.17, with AR yielding significantly higher cylinder values than SR (p<0.001). In the astigmatism vectors, the mean difference for J0 was +0.02 D±0.20, and for J45, +0.02 D±0.20; these differences were not statistically significant. While analysing the refractive astigmatism predictive error (RA-PE), there was a significant difference between AR and SR. The same was observed in the spherical equivalent precision error, with AR and SR differing significantly from each other but not from 0.
Conclusions: AR shows poor correlation with SR in this toric pentafocal IOL. There is a systematic myopic error of 0.52 D in the sphere and 0.68 D in the SE, alongside a systematic error in cylinder power. Correlation in astigmatic components (J0, J45) was also low for the infrared-based autorefractor used in this study.

Sustainability of stainless-steel instruments in cataract surgery: Comparing disposables, recycling, and reusables

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Introduction: This study compares the environmental impact of three scenarios using: disposables, recycling disposables, and reusables in a Dutch health care environment. The aim is to report on the sustainability and environmental impact for stainless-steel instruments used in cataract surgery.
Methods: Building on previously conducted life-cycle assessments, this Life Cycle Assessment employs a cradle-to-cradle material-and energy flow analysis. Three distinct scenarios were evaluated —reuse and recycle— within the framework of circular economy, analysing the system boundaries from mining up until waste management.
Results: The material- and energy flow analyses indicated that the first scenario with disposable tools poses major challenges in achieving circularity and energy efficiency. The recycling scenario reports major carbon emission reduction (>50%) and a shift towards circularity trough reintegrating high quality steel in the European market. The reusable scenario reflects the highest reduction in carbon emissions (>60%), yet yield higher energy consumption and has an imperfect circularity design. This indicates that the reuse and recycle strategies combined perform substantially better.
Conclusion: The study emphasizes the carbon impact of recycling and reusing cataract surgery instruments. The scenario of neither recycling, nor reusing instruments reported to have the highest negative impact on both waste management and carbon emissions. The reusing scenario reported the highest carbon reduction, yet recycling shows to be an opportunity for a circular waste environment. Both arguably just as important. This paper provides nuanced insights to support decision-making for a more sustainable landscape of stainless-steel instruments for cataract surgery.

Six-month performance and safety of an iris-fixated multifocal intraocular lens for presbyopia correction in phakic eyes: A prospective multicenter clinical trial

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Purpose: To evaluate the effectiveness and safety of a novel foldable multifocal iris-fixated phakic IOL (pIOL) for presbyopia correction six months postoperatively.
Methods: This multicenter prospective clinical trial was conducted at seven investigational sites. Presbyopic patients without cataracts undergoing bilateral multifocal pIOL implantation (Ophtec ArtiPlus) were studied. The primary outcomes measured were the visual function at distance, intermediate and near. Other outcome parameters included endothelial cell density (ECD), postoperative refraction, binocular defocus curve, quality of vision, spectacle independence, and patient satisfaction.
Results: Forty-nine patients (98 eyes) were included. At six months postoperatively, the mean binocular uncorrected distance, intermediate, and near visual acuity were −0.05±0.09, −0.02±0.07, and 0.02±0.08 logMAR, respectively. The ECD remained stable (mean value of 2771±289 cells/mm^2). The mean spherical equivalent was −0.41±0.33 diopters (D), with 91.7% eyes having a prediction error within ±1.0D and 71.9% within ±0.5D. The binocular distance-corrected defocus curve showed a visual acuity of 0.20 logMAR or better over a range of +1.0D to −3.5D. Patient satisfaction was high (mean 3.5/4), and 83% of the patients reported spectacle independence. Glare, halos, and starbursts were reported in 47%, 35%, and 96% of patients, respectively, albeit at

non-bothersome levels. Only 2% of patients found halos and starbursts to be very bothersome.
Conclusion: Bilateral ArtiPlus multifocal pIOL implantation demonstrated good visual acuity from distant to near and maintained stable ECD over 6 months. Patients reported high levels of satisfaction and spectacle independence, despite reporting some optical disturbances.

Proteomics reveals mechanisms of delayed keratoconus progression: A study of corneas following two light-activated crosslinking treatments

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Purpose: This study aims to elucidate on changes in biological pathways in rabbit corneas induced by two methods of light-activated corneal stiffening: topical application of riboflavin with dextran (RF-D) or WST11 with dextran (WST-D) followed by ultraviolet A (UVA) or near-infrared (NIR) illumination, respectively.
Methods: Rabbit corneas were mechanically de-epithelialized, then left untreated ($N=3$) or treated with either RF-D/UVA ($N=3$) or WST-D/NIR ($N=3$). After one week, quantitative proteomics was performed on untreated, RF-D/UVA– and WST-D/NIR–treated corneas. Pathway enrichment analysis was performed to identify the biological processes associated with the treatments. To identify the abundance and spatial distribution of lipids in the untreated, WST-D/NIR– and RF-D/UVA–treated corneal stroma, lipid mass spectrometry imaging was performed together with haematoxylin and eosin staining.
Results: Between RF-D/UVA– and WST-D/NIR–treated corneas, 37 and 39 proteins, respectively, were differentially expressed compared to untreated corneas ($p<0.05$). Pathway enrichment analysis showed the effect of RF-D/UVA treatment on cell metabolism and terminal differentiation of keratocytes, while WST-D/NIR modified extracellular matrix regulation and the mitogen-activated protein kinase signalling cascade. When comparing the RF-D/UVA and WST-D/NIR treatment, 74 proteins were differentially expressed, affecting cellular metabolism and respiration, complement activation, the activation of matrix metalloproteinases, and lipoprotein metabolism. The lipid profile for the RF-D/UVA– and WST-D/NIR–treated stromas were similar, whereas differences were observed comparing both treatments to untreated corneal stroma.
Conclusions: Proteomics indicated a metabolic shift from oxidative phosphorylation to glycolysis and hypoxia after RF-D/UVA treatment. In contrast, WST-D/NIR stiffening maintained normal respiration and involved ECM remodelling.

Clinical results of DMEK with and without posturing

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Purpose: To evaluate the clinical results of Descemet membrane endothelial keratoplasty (DMEK) with and without intraoperative or postoperative posturing.
Design: Comparative interventional case series.
Methods: One hundred seventeen (117) consecutive eyes requiring DMEK or Phaco-DMEK for any cause of endothelial failure were included in two chronological cohorts. The first cohort (29 eyes) received strict intra- and 48 h post-operative instructions to remain supine, the subsequent cohort (88 eyes) received no posturing advice. DMEK was typically performed with an intraoperative inferior PI. After unscrolling and centring the graft, the anterior chamber was maximally filled with air, which concluded the procedure. Patients were re-examined after 1 hour at the slit lamp and a 80–90% air fill was assured. Main outcome measures were graft detachment and rebubbling rate. Post-operative complications and long term (6 m) functional results were recorded.
Results: Overall, 43 eyes developed a graft detachment and a rebubbling procedure was performed in 38 of those (38% in the first and 32% in the second cohort). There was no significant difference in the number of detachments (38% vs. 39%, $p=0.95$) or rebubblings (31% vs. 33%, $p=0.78$) between both chronological cohorts. Grafts survived at last follow-up in 113 of 117 eyes (97%), with graft failure occurring twice in both groups (7% vs. 2%, $p=0.24$). Iatrogenic glaucoma (7% vs. 5%, $p=0.62$), IOP steroid responses (10% vs. 11%, $p=0.88$) and CME (7% vs. 7%, $p=0.99$) occurred in equal proportions. At 6 months, overall BCVA was 0.68 vs. 0.78 ($p=0.04$) and ECD 2093 vs. 2001 cells/mm^2 ($p=0.43$) in both chronological cohorts respectively.
Conclusions: Abstaining from intra- or postoperative posturing did not impact the rates of graft detachment, rebubbling, or other postoperative events, even with immediate erect posturing after surgery. This practice can greatly reduce the burden of DMEK surgery in patients, since remaining supine for a prolonged time has considerable medical and practical consequences.

Pluripotent stem cell–derived corneal endothelial differentiation for cell therapies

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Purpose: The differentiation of induced pluripotent stem cells (iPSCs) into corneal endothelial cells (CECs) offers a promising strategy to address the global shortage of donor corneas for transplantation. This approach induces iPSC differentiation through sequential stages, starting with neural crest cells, transitioning to periocular mesenchyme, and finally to CECs. Key signalling pathways, including WNT/β-catenin, FGF, BMP, and TGF-β, regulate each phase. Current differentiation protocols rely on double SMAD inhibition in the TGF-β pathway as the gold standard. However, challenges remain in maintaining lineage specificity, balancing proliferation and differentiation, and avoiding endothelial-to-mesenchymal transition.
Methods: This study investigates a small molecule–based protocol to optimize the differentiation of iPSCs into CEC-like cells. The protocol modulates critical signalling pathways to enhance lineage commitment while using surface coatings to improve differentiation without feeder cells. Differentiation efficiency is assessed using immunofluorescence and flow cytometry to monitor the expression of CEC markers, including ZO-1, ATP1A1, and CD166, while ensuring minimal expression of stem cell and mesenchymal markers such, as SOX2 and CD44.
Results: Preliminary results demonstrate that small molecules provide precise control over pathway modulation, improving differentiation efficiency and reproducibility compared to protein-based factors. Biomimetic surface coatings enhanced cell attachment and monolayer formation, achieving areas of high hexagonal CEC-like cells with characteristic pump markers.
Conclusions: Optimizing iPSC differentiation into CECs using small molecules and surface functionalization offers a scalable, reproducible method for regenerative therapies. This approach could provide an unlimited supply of functional CECs, addressing corneal endothelial dysfunction and reducing reliance on donor tissue worldwide.

Tear fluid proteomics identifies differentially expressed proteins across different stages of Alzheimer's disease and dementia

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Introduction: Alzheimer's disease (AD), the most common cause of dementia, is characterized by progressive neuronal loss and cognitive decline. Tear fluid, an accessible biofluid and collected non-invasively, has emerged as a promising source for AD biomarkers. This study characterized the tear fluid proteome across different stages of AD to identify novel diagnostic biomarkers.
Patients and methods: Tear fluid was collected using Schirmer's strips from 60 participants, including cognitively normal individuals (CN) ($n=32$), and patients with syndrome diagnosis of mild cognitive impairment (MCI) ($n=14$), and dementia (n = 14). Subgroup analyses compared amyloid-beta PET-positive ($n=20$) to PET-negative ($n=8$) patients. Proteins were analysed via label-free liquid chromatography–tandem mass spectrometry. Differentially expressed proteins were identified and analysed for their pathway involvement and tissue origin. Selected proteins were validated using immunoassays.
Results: 703 unique proteins were reliably identified in tear fluid. Seven proteins showed significant differential expression between dementia and CN individuals. In addition to significant upregulation ($p=0.018$), GSS detectability increased with disease severity, from 53.1% in CN to 92.9% in dementia. Two proteins were significantly upregulated in amyloid-beta PET-positive patients compared to PET-negative. Pathway enrichment analysis linked up-and downregulated proteins to neurological disease pathways. Additionally, analysing the tissue origin showed that 19.4% ($n=62$) of up- and downregulated proteins in tear fluid were enriched in the liver.
Conclusion: This study suggests tear fluid as a promising source of biomarkers for AD, with GSS showing diagnostic potential. Future research is essential to confirm

these findings and explore how tear fluid biomarkers can complement existing diagnostic routines.

A cell platform to model Fuchs endothelial corneal dystrophy: Exploring the complexity of a repeat expansion disorder

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Purpose: The CTG18.1 expansion in the transcription factor 4 (*TCF4*) intron 2 is the major cause of Fuchs endothelial corneal dystrophy (FECD), a condition clinically characterized by the loss of corneal endothelial cells. Due to the lack of in vivo models carrying the CTG18.1 expansion and limitations of in vitro platforms, the pathogenic insights characterizing the repeat expansion are yet to be comprehensively outlined. We employed CRISPR/Cas9 to rescue the CTG18.1 expansion in *TCF4* for disease-modelling purposes.
Methods: We isolated peripheral blood mononuclear cells from two FECD patients carrying different CTG18.1 repeat lengths (P1: ~67/83, P2: ~15/130) and one healthy control (Ctrl: ~11/20) and reprogrammed them into induced pluripotent stem cell lines (iPSCs). To rescue the trinucleotide repeat in our patient lines, we applied a dual-step gene-editing approach. First, we generated two Δ/ΔCTG18.1 lines (Δ/ΔP1, Δ/ΔP2), characterized by the biallelic excision of the CTG18.1 expansion. Subsequently, we relied on the homologous direct repair pathway to reintroduce a homozygous fragment of 8 CTG repeats into the Δ/ΔCTG18.1 genetic background, generating two rescued +/+ (CTG)8 lines (+/+ (CTG)8P1, +/+ (CTG)8P2).
Results: Sanger sequencing analysis confirmed the rescue of the Δ/ΔCTG18.1 lines. The pluripotency capacity of +/+ (CTG)8P1 and +/+ (CTG)8P2 was validated by immunocytochemistry of Oct4 and Nanog, and tri-lineage differentiation. Potential CRISPR/Cas9 cleavage activity was excluded by PCR and Sanger sequencing of the predicted off-target sites.
Conclusions: In this work we generated a patient-based cell platform for investigating the contribution of the CTG18.1 expansion to FECD pathogenesis in vitro.

Contact lens fitting after corneal crosslinking for progressive keratoconus

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Purpose: To evaluate contact lens fitting after corneal crosslinking (CXL) for progressive keratoconus.
Methods: This retrospective chart study analysed consecutive data of contact lens (CL) wearers who underwent CXL between 2015 and 2022. Patient demographics, CL type, fitting and parameters (sagittal depth, base curve radius, diameter, lens power) and visual acuity were evaluated before and after CXL.
Results: 110 eyes of 88 patients, 58 OD (53%), were included. Median age at CXL was 26 years (range 14 to 58) and 65 were male (74%). After CXL, 82 eyes (75%) stayed with the same lens type; 4 of 17 (23%) retained rigid gas permeable (RGP) and 75 of 83 (90%) retained scleral lenses (SL). Four eyes with large diameter SL were fitted into mini-SL. Of 71 eyes with the same SL type before and after CXL, 59% showed stable or better visual acuity, 47% received lenses with the same spherical power and 79% with the same cylindrical power. All SL parameters were stable ($p > 0.05$). Habitual SL fitting was assessed at median 12 weeks (range 4–40) post-CXL and was good in 56%, acceptable in 23%, unacceptable in 1.4% and unknown in 20%.
Conclusion: Most SL wearers retained the same lens type post-CXL, while RGP wearers changed lens type in almost three quarters of cases. Despite a general flattening of the cornea after CXL, SL fitting parameters remained stable. Unacceptable fitting after CXL with habitual SL was rare; SL can be worn safely after CXL, but evaluation of lens power is recommended.

Asynchronous virtual clinic evaluation in patients with a broad range of clinically stable corneal and anterior segment pathology: A report on efficacy

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Purpose: Reporting the efficacy of asynchronous virtual clinic evaluation in patients with a broad range of clinically stable corneal and anterior segment pathology.
Methods: Patients with clinically stable corneal or non-uveitic anterior segment pathology were evaluated in an asynchronous virtual clinic setting instead of their regular annual clinical check-up. During the visit history, visual acuity, intraocular pressure were evaluated and patients underwent macular OCT and either corneal tomography (Pentacam) or slit lamp photography. Results were evaluated and communicated a few days later. Any patient with complaints, signs of new disease or disease

progression was summoned for a physical re-evaluation by an ophthalmologist. The number of patients that required physical re-evaluation and the fraction of patients in which re-evaluation changed medical treatment were reported.
Results: A total of 264 patients were evaluated, of which 48 (18.2%) were summoned for a physical re-evaluation. A total of 12 (4.5%) patients reported complaints and an additional 27 (10.2%) patients showed signs suspect for new or progression of disease. Imaging was not interpretable in 2 patients and in 7 patients one of the investigations was missing. In 39% of the patients that were summoned for a regular ophthalmologist visit, disease progression or a new disease was found such that it changed medical treatment.
Conclusions: Asynchronous virtual clinic evaluation of patients with corneal pathology provides a valuable opportunity to efficiently evaluate patients with stable corneal and non-uveitic anterior segment pathology.

The NTS cornea-dashboard: A national transparent up-to-date quality registry for corneal transplant surgery

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Background: Corneal transplantation is a critical procedure for restoring vision. Since 2007, the Dutch Transplant Foundation (NTS) has hosted the Dutch Organ Transplant Registry (NOTR). This provides a comprehensive obligatory web-based standardized data collection on outcomes of corneal transplant surgery in The Netherlands.
Objectives: This presentation highlights key features of the Cornea-dashboard, a dashboard on national level. It highlights its utility in improving surgical planning, benchmarking outcomes, ensuring data completeness, and reporting on developments in practice patterns.
Methods: The Cornea-dashboard visualizes the data collected in the NOTR, enabling intuitive access and insights. It is programmed in Microsoft Power BI and provides data (planning of) corneal transplant procedures, patient demographics, and post-operative outcomes across multiple centers. Follow-up data on graft survival, rejection rates, and complications are recorded and analysed. Quantitative data from the registry can be used to illustrate the platform's impact for the relevant stakeholders.
Results: To date 26.221 surgeries are recorded. An average increase of 55 recorded surgeries each year can be appreciated, next to an overall transition from predominantly PK (2007:75%) to DSEK (2014: 57%) to DMEK (2024: 61%). Data-completion is 94% after surgery, and 77% resp. 64% at one and 2 year follow-up.
Conclusions: The Cornea-dashboard offers a comprehensive tool to enhance transparency, data-driven decision-making, and long-term patient care. By providing insights into planning, surgical outcomes and facilitating benchmarking, it supports both clinical and operational improvements in corneal transplantation. Assuring a complete dataset is instrumental to achieve these goals.

A new tool for improved automated analysis of corneal endothelial cell using artificial intelligence

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Introduction: Presentation of our AI-assisted platform designed for automated segmentation and quantification of corneal endothelial images in healthy- and pathological eyes, named CELDA (Corneal Endothelium Large Dataset Analysis).
Methods: CELDA employs deep learning models for automatic segmentation of a region-of-interest (ROI), i.e. areas with measureable endothelial cells, in specular microscopy images. The platform computes cell density (ECD), hexagonality (HEX), and variant in cell size (CV), in single images and entire datasets. Additional features include segmentation visualization, summary statistics, and quality control labeling. Users can also export features and segmented images for further analysis. CELDA was evaluated by a multidisciplinary team using data from healthy controls and glaucoma patients. Results were assessed through visual inspection and compared to outputs from the built-in software.
Results: The AI method used in CELDA was evaluated with the following results: in corneas with guttata our method achieved a 98.4% success rate for ECD, CV, and HEX, compared to 71.5% for ECD and CV, and 30.5% for HEX with the built-in software. The errors for our measurements were 2.5% (ECD), 5.7% (CV, HEX), versus 7.5% (ECD), 17.5% (CV), 18.3% (HEX) for the built-in software ($p<0.0001$, paired Wilcoxon). (ref).
The multidisciplinary team evaluated CELDA as a user-friendly and were capable of applying the tool in order to assess the endothelial parameters.
Conclusions: Our AI-based tool outperforms built-in software for automated corneal endothelial analysis, offering more accurate and reliable results. The user-friendly interface enhances its usability, making it a valuable asset for research applications.

Prescription patterns in descemet membrane endothelial keratoplasty (DMEK): A European survey

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Purpose: To assess current prescription patterns in Descemet Membrane Endothelial Keratoplasty (DMEK) in Europe.
Methods: The steering group of the European Cornea and Cell Transplantation Registry (ECCTR) and the European Vision Institute Clinical Research Network (EVICR.net) sent an electronic survey to 16 national societies. Questions regarding prescription patterns related to routine and high-risk DMEK, and management of graft rejection and steroid-induced ocular hypertension were included. Data were analysed using descriptive statistics.
Results: Responses from 136 surgeons revealed that medication protocols mainly came from departmental protocols (54%) or personal experiences (48%) rather than national guidelines (22%) (multiple answers were allowed). Preoperatively, most surgeons (60%) did not prescribe any medication; others prescribed topical hyperosmolar solution (21%), antibiotics (15%), and steroids (12%). During surgery, antibiotics (72%), miotics (66%), and steroids (59%) were common. Postoperatively, all surgeons prescribed steroids, with antibiotics (92%) and lubricants (38%). Steroids were typically tapered to once daily after six months (46%) and discontinued after varying durations (65%). Dexamethasone was the preferred steroid. For high-risk DMEK, the most common approach was adding another topical (30%) or systemic immunosuppressive drug (24%), such as cyclosporine or mycophenolate. For graft rejection, most respondents increased topical steroid frequency (85%) or added (peri) bulbar steroid injections (42%). For steroid-induced ocular hypertension, the first approach was switching to a lower-potency steroid (40%) or reducing steroid frequency (43%).
Conclusions: Current prescription patterns in routine and high-risk DMEK vary significantly before, during, and after surgery, reflecting the lack of standardized guidelines.

Eye in the sky: Posture and movement in the first 24 hours as risk factors for transplant detachment after DMEK

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Introduction: Postoperative graft detachment with corresponding need to rebubbling is the most frequent complication after DMEK surgery. There is no consensus about restrictions in posture and movements after the operation. With this study we analyse the effect of posture and movement of the patient in the first 24 hours after the operation on the detachment rate of the DMEK transplant.
Methods: An inertial measurement unit sensor, able to detect acceleration and angular velocity was attached to the patient forehead during the first 24 hours after the operation. Results of 61 Fuchs patients undergoing DMEK transplantation were included in the analysis. Movement and head inclination parameters were constructed and tested with multivariate logistic regression against (1) rebubbling and (2) graft detachment and/or rebubbling.
Results: Of the 61 patients, 6 required rebubbling, while 15 were categorized within the detachment and/or rebubbling group. No significant correlation was observed between movement and rebubbling. However, a significant association ($p<0.05$) was identified between mean head inclination and rebubbling. A more supine position was favourable for graft attachment. Additionally, no significant relationship was found between the movement and head inclination parameters and the detachment and/or rebubbling group.

Conclusion: The posture instructions after the DMEK operation might be less strict with regard to movement, however a supine position during the first 24 hours is favourable to prevent graft detachment.

Influence of unfolding time of DMEK roll on ECD and re-bubbling/re-grafting rate postoperatively, a retrospective study of 112 consecutive cases

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Purpose: To analyse the possible effect of unfolding time of DMEK roll on postoperative re-bubbling/ re-grafting rate and endothelial cell loss.
Methods: Outcomes of 112 patients were evaluated, including 109 with FECD and 3 eyes with ICE syndrome. Average age was 69.8 years. 75 eye's (67%) underwent single DMEK and 37 eye's (33%) a triple DMEK. Average preoperative VA was 0.35, average of ECD was 2746 mm^2. We defined 3 categories, category I: unfolding time<5 minutes, category II: 5>min<10 min and category III: >10 minutes.
Primary outcomes were correlation between unfolding time and re-bubbling & re-grafting rate and endothelial cell loss postoperatively. Secondary outcome was postoperative VA.
Results: Unfolding time in cat I was achieved in 67% of the eye's followed by cat II in 18.2% and cat III in 6.1%.
No significant correlation between cat I & II regarding re-bubbling rate (p=0.60). We found well significantly more re-grafting in cat III comparing with cat I & II (p=0.02, chi-square test).
We found no statistical correlation between cat I and II and postoperative endothelial cell loss at 1 year (p=0.289). Also, no significant correlation could be found between cat I and II and postop VA at 1 year (p=0.219).
Conclusion: Unfolding time<10 min could decrease the rate of re-grafting comparing to unfolding time>10 min. ECD was slightly higher and VA was slightly better in cat I comparing to cat II, but correlation was not significant.

Reliability and agreement of specular microscopes in corneal endothelial cell density: Comparing built-in vs. AI segmentation

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Purpose: To evaluate the reliability and agreement of three specular microscopes in measuring corneal endothelial cell density (ECD) and compare their built-in segmentation methods with a new AI-based approach.
Methods: We analysed healthy corneas of 60 glaucoma patients with Topcon SP-1P, Konan CellChek-20, and Tomey EM-4000. One eye per patient was photographed on all devices, capturing three images at the central (C) and temporal superior (TS) cornea. ECD was determined using each device's built-in software and an in-house algorithm. We assessed test–retest reliability with two-way mixed, single measures, absolute agreement intraclass correlation coefficients (ICC), and inter-device agreement via variance components-derived ICCs from a linear mixed-effects model.
Results: The mean age was 69±9 years (40% female). Thirty-eight participants (63%) had prior phacoemulsification, one had trabeculectomy, and one had vitrectomy. Test–retest ICCs at C were 0.951 (Topcon), 0.959 (Konan), and 0.846 (Tomey) with built-in software, and 0.978, 0.973, and 0.983 with AI. At TS, ICCs were 0.921 (Topcon), 0.904 (Konan), and 0.894 (Tomey) with built-in software, and 0.943, 0.987, and 0.973 with AI.
Inter-device agreement ICCs at C were 0.928 (Topcon-Konan), 0.937 (Topcon-Tomey), and 0.916 (Konan-Tomey) with built-in software, and 0.941, 0.962, and 0.947 with AI. At TS, ICCs were 0.797 (Topcon-Konan), 0.850 (Topcon-Tomey), and 0.826 (Konan-Tomey) with built-in software, and 0.886, 0.943, and 0.916 with AI.
Conclusions: Our findings indicate that, while the three devices demonstrated generally strong reliability and agreement with built-in software, the AI model appears to offer greater consistency in calculating ECD. Further analyses to evaluate these improvements are underway.

Visual improvement of coma correction in scleral lenses for keratoconus patients

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Purpose: To evaluate the visual effect of coma correction in scleral lenses (wfSL) compared to spherocylindrical scleral lenses (stdSL) for keratoconus (KC) patients.
Methods: Fifty-two eyes of fifty-two KC patients were reviewed. Participants were wearing or were fitted with a stdSL and a wfSL. In a wfSL only coma correction was added by an optometrist, based on readings of higher order aberrations (HOA's) and visual performance (VP). Visual acuity (VA) and HOA's were evaluated in both lens types. Higher order aberrations were determined by aberrometry (I-Trace, Tracey Technology, Texas) and evaluated over a pupil scan of 5.0 mm. After 1 month of wear, 12 patients were interviewed to evaluated subjective visual performance of stdSL versus wfSL. For this purpose we used a scale of 1–5 were 1=strongly deteriorated, 2=deteriorated, 3=same result, 4=improved, 5=strongly improved.
Results: Mean VA improved from VA 0.71 in stdSL to VA 1.04 in wfSL ($p<0.001$). HOA's reduced in a 5.0 mm scan from 0.868 μ in stdSL to 0.632 μ in wfSL. Twelve participants noticed VP for daily live activities (4.4) and improved binocular vision (4.3), and also mentioned less photophobia (3.8) and hinderance of halo's and starburst (3.9).
Conclusions: wfSL with only coma addition for KC patients can improve VA compared to stdSL, and can also have an improving effect on hinderance of photophobia, halos and starburst.

5 years experience with an endothelial keratoprosthesis

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Purpose: To report 5 years results of implantation of the EndoArt (Eye Yon Medical, Israel) in terms of safety, device adherence, corneal thickness, central cornea clarity, adverse effects, and visual acuity.
Methods: Retrospective case series. The EndoArt is a prosthetic dome shaped acrylic implant of 6.5 mm diameter, that functions as a physical barrier preventing fluid uptake into the cornea stroma leading to a new equilibrium & decrease in corneal thickness. The EndoArt was implanted in 12 eyes of 12 patients for indications of corneal edema. In 10/12 eyes the indication was tectonic, because of complex posterior segment pathology.
Results: Follow-up was 6–62 months. In 11/12 eyes the EndoArt remained adherent throughout the follow up period. In 9/12 eyes a temporary corneal suture was used to aid initial adherence. No perioperative or post operative complications occurred, no pathological thinning of the cornea was seen. In 1 patient a retinal detachment occurred at 44 months. In 8/12 visual acuity improved. Central corneal thickness on OCT decreased with a mean of 26.8%. In 1 patient no effect was seen. In 1 patient the Endoart failed to adhere.
Conclusions: The EndoArt remained in situ in 11 patients in this cohort. The EndoArt implantation reduced corneal thickness. In 8/12 patients visual acuity improved. No complications related to the implant were seen. Suturing the device led to improved adherence. More follow up in more patients is needed to elucidate the exact indications for the use of an artificial corneal endothelial device.

Dupilumab Associated Ocular Surface Disease – an update on ocular adverse events during systemic biologicals use in eczema treatment

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Purpose: to describe the epidemiology, clinical features and treatment options of ocular adverse events of Dupilumab and other biologicals in eczema treatment.
Methods: data were collected of eczema patients at multidisciplinary allergy outpatient departments performed by ophthalmology and dermatology.
Tear fluid, blood and impression cytology samples were collected and tested for inflammatory markers and drug levels of dupilumab and tralokinumab.
Clinical signs and symptoms including evaluation of ophthalmic treatment were described by the UTrecht OPhthalmic Inflammatory and Allergic ocular surface disease scoring system (UTOPIA).
Results: 45% of eczema patients treated by dupilumab develop a level of Dupilumab-associated ocular surface disease (DAOSD). 33% of these develop severe disease. Up to 50% need chronic use of steroids for treatment. Less than 10% of ocular surface patients needed withdrawal of dupilumab. In less than 10% limbal stem cell damage was seen.
Correlations were seen between more severe ocular disease and higher dupilumab levels in tear fluid, a history of any ocular disease, decreased number of goblet cells in conjunctiva and elevated TARC levels in tear fluid.
Newer biologicals show similar side effects, especially in patients with former DAOSD complaints on dupilumab.
Conclusions: dupilumab can cause severe ocular surface disease and risk for limbal stem cell deficiency. Close monitoring and treatment by ophthalmologists may be necessary.

Polyhexanide 0.8 mg/mL to treat *Acanthamoeba* keratitis: multi-center, real-world experience in The Netherlands

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Purpose: *Acanthamoeba* keratitis (AK) is a rare severe disease with high patient burden, difficult diagnosis and no internationally agreed standard-of-care treatment. In August 2024, polihexanide 0.8 mg/mL (PHMB 0.08%) was approved in the European Union for AK treatment in patients aged ≥12 years. We aim to describe the first experience of using PHMB 0.08% for AK treatment in The Netherlands.
Methods: Multi-center, retrospective case series of AK patients treated with topical PHMB 0.08% (October 2022–December 2024, compassionate use). We assessed baseline characteristics and treatment success (i.e., no clinical signs of active infection/inflammation after epithelial closure).
Results: Eight AK patients were included (mean [range] age: 52.1 [18–77] years; 5/8 female; 7/8 contact lens wearer), with 1 having stage I, 6 stage II, and 1 stage III disease. All patients received prior anti-amoebic treatment (8/8: chlorhexidine 0.02%; 6/8: propamidine 0.1%), and 5/8 patients prior corticosteroids. Time from symptom onset to PHMB 0.08% treatment was ≥21 days in all patients (range: 21–1201 days). 6/8 patients achieved treatment success, 1/8 is undergoing treatment, and 1/8 has data pending yet. PHMB 0.08% treatment regimen and use of other topical treatments in some patients were at physicians' discretion per clinical practice. No PHMB 0.08%-related safety concerns were reported.
Conclusions: PHMB 0.08% was successfully used in daily clinical practice in pre-treated, complex AK cases with long treatment start delay. More experience is needed in daily practice with PHMB 0.08% monotherapy for AK according to the published clinical management protocol (Dart et al. Ophthalmology 2024;131:277–287).

Impact of cartridge size on DMEK graft implantation: optimizing technique for donor age and tissue characteristics

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Introduction: Efficient DMEK graft implantation depends on selecting the appropriate cartridge size, tailored to the donor tissue's properties. Younger donor tissues are tighter, making handling challenging, while older, looser grafts are less demanding. This study evaluates the performance of 14Ga and 16Ga cartridges in various donor scenarios, emphasizing graft orientation, implantation control, and procedural outcomes.
Materials and Methods: Video recordings of DMEK procedures involving four donor tissues were analysed. A 16Ga cartridge was used for a 72-year-old loose double-roll graft and a 45-year-old donor graft, while a 14Ga cartridge was utilized for a tight double-roll graft from a 31-year-old donor and a loose double-roll graft from a 73-year-old donor. Outcomes assessed included graft orientation during implantation, control over the graft, unfolding process, and intraoperative complications.
Results: The 16Ga cartridge maintained graft orientation effectively and enabled smooth implantation, particularly for younger, tighter grafts. However, in the case of a loose graft, the narrow cartridge caused the tissue to eject at high speed, landing upside down, necessitating additional manoeuvres. The 14Ga cartridge allowed freer graft movement but resulted in reduced orientation control with younger donor tissue. Spontaneous bleeding during one implantation was successfully managed, but fibrin formation remains a risk if bleeding is not cleared promptly.
Conclusion: Cartridge selection should align with donor age and tissue characteristics. The 16Ga cartridge is ideal for tighter grafts from younger donors, while the 14Ga cartridge is more suited for looser grafts from older donors, minimizing complications and improving surgical outcomes.

DMEK graft preparation success rate in donor corneas with post-phacoemulsification scars

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Purpose: To evaluate the success rate of DMEK graft preparation in donor corneas with post- phacoemulsification scars.
Methods: Retrospective study on data from DMEK graft preparations in October and November 2024 in our Eye bank. DMEK grafts were dissected from 67 corneas with peripheral post-cataract surgery scars by four technicians, using the 'no touch' technique to harvest an isolated Descemet roll. Graft quality was evaluated with inverted light microscopy.
Results: Successful graft preparation was obtained in 81% of dissections (n = 54). The remaining corneas were rejected due to inadequate quality in 16% (n = 11), and dissection failure occurring in 3% (n = 2). Success rate of graft preparation using corneas without post-cataract scars was 90%.
Conclusion: Donor corneas with peripheral post-cataract scars can be successfully used for DMEK graft

preparation, potentially increasing the pool of usable donor tissue.

Determining the variability associated with visual acuity and refractive error measurements: A systematic review

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Introduction: Accurately measuring visual acuity (VA) and refractive error (RE) is fundamental to clinical practice, although no scientific consensus has been reached with respect to variability. This systematic review examines the variability of measuring visual acuity (VA) and refractive error (RE).
Methods: This systematic review was performed in accordance with PRISMA guidelines and was registered at PROSPERO (IDCRD42024554663). We searched the Medline, PubMed, and Embase databases. The Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool was used to assess the risk of bias (RoB). We included studies that directly compared VA or RE and that reported the Limits of Agreement (LoA) and mean difference.
Results: Our analysis included 13 and 6 studies that reported VA and RE, respectively. From these 19 studies, only 6 (32%) scored a low RoB in at least three out of four QUADAS-2 domains. In 27 out of 38 subgroups (71%), LoA exceeded the generally accepted ranges for VA or RE (namely, ±0.15 LogMAR and±0.5 D, respectively). Based on our analysis, the summarized mean LoA values were±0.20 LogMAR (95%CI 0.17;0.23) and±0.70 D (95%CI 0.50;0.89) for VA and RE, respectively.
Conclusion: We found that measuring both VA and RE have a high variability, exceeding currently accepted limits. Thus, the current limits are either unrealistic or reflect studies with methodological weaknesses. Rigorous studies are therefore needed to more accurately estimate the acceptable limits of variability. For now, we propose using the mean LoA values obtained in our study as a provisional frame of reference.

Henri Parinaud & Xavier Galezowski, namegivers of a disease entity more than a syndrome

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Purpose: To reconstruct the etymologic origin of (Galezowski-)Parinaud's oculoglandular syndrome.
Methods: Literature study and online archival research.
Results: On February fifth, 1889, at the meeting of la Société d'Ophthalmologie de Paris in l'Hôtel des Sociétés Savantes, Henri Parinaud presented three cases with unilateral granulomatous conjunctivitis with ipsilateral regional lymphadenopathy accompanied by fever with irregular shiverings for a prolonged period, however not severely affecting the general wellbeing. He presumed the conjunctivitis being infectious and transmitted by animals. Xavier Galezowski reacted that he himself had encountered about twenty similar cases. Since then there has been controversy whether the pattern of symptoms in the patients described by Parinaud and Galezowski compounded merely a nonspecific syndrome or defined a single clinical entity. Parinaud believed the latter, but did not provide a specific pathomechanism. Because a wide spectrum of possible infectious causes were attributed, the combination of symptoms became eponymously known as (Galezowski-)Parinaud's oculoglandular syndrome. After almost a century finally the most common infectious agent was assessed: Bartonella henselae. Nowadays B. henselae is even seen as the single most important cause of Parinaud's oculoglandular syndrome and therefore it was recently suggested that the term “Parinaud's oculoglandular syndrome” is to be replaced by “Parinaud's oculoglandular disease” and only used in cases with the clinical findings in combination with a Bartonella infection.
Conclusion: It took more than a century to confirm that Parinaud and Galezowski identified a new disease entity.

Synthetic retinal images for the diagnosis of Alzheimer's disease using generative artificial intelligence

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Introduction: Alzheimer's disease (AD) is characterized by accumulation of Amyloid Beta (Aβ) plaques and phosphorylated tau tangles. Both have also been shown to be present in the retina, making retinal imaging a promising screening method for AD. Advances in artificial intelligence (AI) have allowed for the development of convolutional neural networks (CNN) based AD prediction models. This study assessed denoising diffusion probabilistic models (DDPMs) to generate synthetic multimodal retinal images to investigate its potential in discriminating Aβ+from Aβ- participants in CNN.
Materials and Methods: Fundus autofluorescence (FAF), optical coherence tomography (OCT) and optical coherence tomography angiography (OCTA) were performed on 167 participants (328 eyes) to train DDPMs to generate synthetic images. Among these participants, 35%

were Aβ+. For each Aβ class, 1000 synthetic images were generated, which were then used to pretrain CNNs. Unimodal and multimodal classifiers were trained to predict Aβ status.
Results: DDPMs generated diverse, realistic retinal images with minimal memorization. The best performing model was multimodal, incorporating metadata trained on real images, and achieved an AUC of 0.729. The best unimodal model was trained on synthetic OCTA macular images and achieved an AUC 0.647. Class activation maps (CAMs) were used to identify retinal features associated with Aβ status and showed significant correlations with the foveal avascular zone and the superior temporal region.
Conclusion: This study showed that synthetic data generated through DDPMs can improve Aβ status prediction using multimodal retinal imaging. This study serves as proof of concept for leveraging generative AI in AD prediction using multimodal retinal images.

Blended learning during clinical rotations

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Purpose: Medical students often struggle to identify and study relevant information due to the vast amount of available data. This is especially true in ophthalmology, where students start clinical rotations with limited knowledge and face restricted clinical exposure. A modular blended learning program was developed to enhance knowledge and optimize clinical exposure, adaptable for any medical specialty.
Methods: The program combines interactive education and self-study modules, including an escape room and a virtual reality simulator to reactivate prior knowledge and teach examination techniques. During rotations, 'just-in-time learning' slots are allocated for studying modules covering basic knowledge, clinical preparation, in-depth study, and formative assessments. Content is delivered through e-learnings, recorded lectures, surgery videos, knowledge clips, animations, website passages, and textbook readings.
Results: Post-rotation surveys from 82 students using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) rated the program 4.6 (SD 0.6) for knowledge acquisition and 4.4 (SD 0.8) for clinical preparation. Students reported a steep learning curve and an improved ability to distinguish between normal and abnormal situations. The targeted preparation boosted their confidence, leading to more questions and proactive behaviour in clinical settings.
Conclusions: Students are highly enthusiastic about the blended learning program. The combination of clinical exposure and self-study results in a steep learning curve, enabling students to complete the program with strong clinical knowledge and making optimal use of clinical exposure opportunities. Experiences and implementation tips for this modular 'just-in-time learning' approach will be shared.

Patient satisfaction and efficiency of an asynchronous virtual ophthalmology clinic

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Purpose: Healthcare is under tremendous pressure due to rising healthcare costs and increasing waiting lists. This is forcing ophthalmologists to innovate in order to be able to continue providing high-quality healthcare. An asynchronous (data collection and evaluation at different points in time) virtual clinic might be a more efficient alternative to a classic outpatient-clinic visit.
Methods: In 127 patients who visited an asynchronous virtual clinic we recorded how much time was spent by the resident, the ophthalmic technicians, and the supervising ophthalmologist. Furthermore, patient satisfaction was measured using a 4-point scale ranging from very unhappy to very happy.
Results: On average a resident spent 5.5 minutes per patient, which includes writing the medical findings letter and making a follow-up appointment. The supervising ophthalmologist spent 42 seconds per supervised patient, with supervision required in 44 patients. Per half-day virtual clinic, 21 patients were assessed by three ophthalmic technicians. 98% of patients were happy or very happy.
Conclusions: Asynchronous ophthalmology clinics were shown to be time efficient and might play an important role in reducing healthcare costs and waiting lists.

Virtual clinic long-term (1.5 years) follow-up

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Purpose: To counter the increasing access times for ophthalmic consultations, a virtual clinic to replace regular long-term ophthalmic outpatient consultations was started in 2023. In this study we investigated the long-term (1.5 years) outcome of these virtual clinic visits.
Methods: From February to July 2023, 171 patients with a long-term (>6 months) appointment (for either the retina or glaucoma service) were seen in a virtual clinic setting. In the virtual clinic, standardized diagnostic measurements (visual acuity, IOP, OCT, and Optos fundus photography) were performed by a technician. These measurements were reviewed on a separate occasion by a physician. After 1.5 years, the ophthalmic charts of these patients were reviewed.
Results: At 1.5 years follow-up there were no indications of missed diagnoses. Nine patients had new ophthalmic disorders during the follow-up period, however all of these disorders developed more than 6 months after their virtual clinic visit.
Conclusions: After long-term follow-up an ophthalmic virtual clinic seems to provide a good quality of care.

Evaluating costs and utilities of myopia across treatment scenarios in The Netherlands

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Purpose: The prevalence of myopia is rising with associated complications expected to impose a significant (financial) burden on healthcare systems and society. This health-economic evaluation investigates the impact of myopia control on costs and quality of life in the Dutch population.
Methods: A state-transition model simulated per-diopter risks of myopic complications from a societal perspective. Individuals with myopia entered the model at adulthood. Age-specific, per-diopter risks of complications were derived by combining emmetrope incidence rates with per-diopter prevalence increases. Hypothetical myopia control interventions were modelled based on the current distribution of myopia in The Netherlands. Cost-utility outcomes were compared to the Dutch willingness-to-pay (WTP) threshold of €20 000/QALY.
Results: Average discounted lifetime costs and QALYs ranged from €15 790 and 37.1 QALYs at −1 D, to €31 676 and 33.9 QALYs at −10 D. Across the current myopia population, costs and benefits averaged €17 055 and 36.7 QALYs per person. Preventing myopia by 1 D across all individuals yielded an average gain of 0.11 QALY per person, requiring intervention costs below €2613 to be cost-effective at the WTP-threshold. Targeting individuals with high myopia (≤ −6 D) increased QALY gains to 0.49 per person, permitting intervention costs of up to €11 529.
Conclusions: Expenditures and QALY losses increase exponentially with higher degrees of myopia. At −10 D, individuals lose an average of 3.2 QALYs due to myopia. Interventions achieving a + 1 D shift provide QALY gains across all degrees of myopia, but focusing on high myopia offers the greatest cost-effectiveness potential.

Mohs micrographic surgery in periocular tumours: A comparison between two decades

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Introduction: Skin cancer is one of the most common cancers worldwide and the incidence is rising. Treatment increasingly consists of Mohs micrographic surgery (MMS) in cases of periocular involvement, to minimize morbidity and spare eyelid and lacrimal system function. An overview of the patients treated in our multidisciplinary MMC programme is given in this retrospective study in which two time spans from different decades are compared.
Methods: Patients with periocular tumours were referred by a general practitioner, dermatologist, or ophthalmologist to the Erasmus MC Hospital and treated in the multidisciplinary MMS program in the time period 2007–2009 and 2021–2023. All patients were treated by both the ophthalmologist and dermatologist. Clinical and histopathological data were collected and an overview was made.
Results: During the time periods 2007–2009 and 2021–2023, 63 and 129 patients were treated respectively. More females were treated in both groups. Most tumours were basal cell carcinomas in both time periods (90% and 74%; $p < 0.05$) and the majority of these comprises of the nodular type (60% and 62%; $p > 0.05$). More advanced reconstruction techniques such as Hughes flap were used during 2021–2023 compared to the earlier time period due to the complexity of the cases.
Conclusion: There is an increase in patients at the Erasmus MC that underwent MMS for periocular skin tumours in a multidisciplinary programme, when comparing 2007–2009 and 2021–2023. More difficulty to achieve free resection margins and more advanced reconstructions suggest the complexity of the periocular cases has increased in the later time period.

Efficacy of tele-optometric consultations in managing ocular complaints: A 15-month study of 1080 cases

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Purpose: The purpose of this research was to evaluate the efficacy of tele-optometric consultations in managing ocular complaints. By integrating telemedicine technologies with optometric care in optician shops, we aimed to determine the accuracy and clinical outcomes of remote diagnostics and management.
Methods: A total of 1080 patients presenting with various ocular complaints at optician shops were included in this study, conducted over a 15-month period. Comprehensive anterior segment imaging was employed to assess the cornea, anterior chamber, and lens. For

posterior segment evaluation, fundus photography and optical coherence tomography (OCT) were utilized. All cases were analysed by qualified optometrists, with supervision by an ophthalmologist to ensure accuracy and quality in diagnosis.
Results: A wide range of diagnoses was established through tele-optometric consultations. The top three diagnoses were keratitis sicca, cataract, and glaucoma suspect. In 82.8% of cases, a tele-optometric consultation was sufficient to establish a diagnosis. In 9.4% of cases, a referral to an ophthalmologist was necessary, and in 6.6% of cases, a physical optometric examination was planned. The average waiting time for patients to receive a tele-optometric consultation was 3 days. Optometrists conducted an average of 6 consultations per hour. Patient satisfaction, measured by the Net Promoter Score, was rated at 8.7.
Conclusions: Tele-optometry is a safe and effective alternative to physical optometry. Its increased accessibility allows patients to receive faster ophthalmological care while maintaining high diagnostic accuracy and patient satisfaction.

The ocular phenotypic spectrum in Stickler Syndrome: an analysis of visual acuity, refractive error and axial length progression

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Purpose: Stickler syndrome is an inherited connective tissue disorder that confers a significant risk of severe ocular disease and complications, including high myopia and retinal detachments (RD). Limited research exists on spherical equivalent (SER) and axial length (AL) progression in these patients. More insight in ocular growth is key to improving myopia control and patient care.
Methods: Longitudinal data on SER, AL, best-corrected visual acuity (BCVA) and myopia control of genetically confirmed Stickler patients was collected in multiple ophthalmogenetic centers in The Netherlands. The BCVA of the better-seeing eye, mean SER and AL were determined at ages 4, 10, and 16 years. The number of RD, cataracts and proportion of patients treated with atropine and other forms of myopia control during follow-up was reported.
Results: To date, 26 patients with Stickler syndrome were included (15/26 males, age range: 0–39 years). The disease was caused by COL2A1 (n = 24), COL11A1 (n = 1), or COL9A1 (n = 1) variants.
Mean SER/AL/BCVA at 4 years was −6.58D (SD 5.8)/25.02 mm (SD 1.30)/ 0.65 (SD 0.13), −6.86D (5.92SD)/25.74 mm (SD 1.03)/0.80 (SD 0.28) at 10 years and − 7.32D (8.95SD)/27.56 mm (SD 0.13)/ 0.70 (SD 0.14) at 16 years. 7/26 received atropine therapy: 0.01% (n = 1), 0.05% (n = 4), 0.5% (n = 1) and both 0.01%, 0.05%, and 0.1% (n = 1). During follow-up, 3 patients developed RD and 1 cataract.
Conclusions: Our data confirm the link between high myopia, axial length, and Stickler syndrome, despite a highly variable phenotype. Next, we aim to create Stickler-specific eye growth curves to improve myopia prognosis and patient care.

Randomized clinical trial to evaluate the efficacy of acetazolamide for the treatment of cystoid fluid collections in X-linked retinoschisis: The AXIS trial

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Purpose: This RCT evaluates the efficacy of oral acetazolamide in reducing cystoid fluid collections (CFC) and improving visual function in X-linked retinoschisis (XLRS) patients.
Methods: This investigator-initiated, single-center RCT, conducted at the Amsterdam University Medical Center, involved seven visits over 32 weeks. XLRS patients older than 12 years with foveal CFC were randomly assigned (1:1) to either acetazolamide treatment or active monitoring. The primary endpoint was the change in central subfield thickness (CST) on SD-OCT from baseline to the final visit at week 32. Secondary endpoints were the change in best-corrected visual acuity, low-luminance visual acuity, mean retinal sensitivity on microperimetry, and vision-related quality of life using the Michigan Retinal Degeneration Questionnaire from baseline to 32 weeks.
Results: Between January 20, 2023, and February 2, 2024, we enrolled 19 patients to receive acetazolamide (n = 10, 17 eyes) or active monitoring (n = 9, 16 eyes). The mean change in CST was −34.0 ± 167.1 µm in the treatment group and − 42.1 ± 102.2 µm in the control group (p = 0.76). No significant changes were observed for the secondary outcome measures. On an individual level,

five participants in the treatment group experienced an improvement in BCVA of more than 7 letters in one eye, while one participant showed such improvement in both eyes. In contrast, no patients in the control group achieved this level of improvement during follow-up.
Conclusions: We do not recommend the universal use of acetazolamide for managing CFC in XLRS patients. However, a subset of patients may still derive structural and functional short-term benefits from this therapy.

Two-year follow-up of Dutch patients with *RPE65*-associated inherited retinal dystrophy: Restoration of rod-mediated scotopic vision following treatment with voretigene neparvovec

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Purpose: To examine changes in scotopic vision in patients with *RPE65*-associated inherited retinal dystrophy following treatment with voretigene neparvovec (VN).
Methods: Scotopic full-field threshold stimulus (FST) testing was the primary endpoint, and was analysed in a multi-center, prospective, observational study of patients treated with VN. FST sensitivity served as a biomarker for retinal function, with improvements defined as gains of at least 1 log cd·s/m^2 and a blue-red difference of at least 2 log cd·s/m^2 indicating rod-driven scotopic vision. Treated eyes were grouped based on FST changes, and trends in best-corrected visual acuity (BCVA), low-luminance visual acuity (LLVA), visual field area measured by semi-kinetic perimetry (SKP), and retinal pigment epithelium (RPE) integrity were analysed.
Results: Rod-mediated scotopic vision was partially restored in 77% of eyes with persistent FST sensitivity improvement by month 1, and up to 69% by 24 months. An intact RPE area of at least 3 optic disc diameters before treatment predicted both the restoration and maintenance of rod-mediated scotopic vision.
Conclusions: RPE integrity on retinal imaging may serve as a potential biomarker for predicting visual outcomes after treatment with VN.

Two-year follow-up of Dutch patients with *RPE65*-associated inherited retinal dystrophy: Clinical outcomes following treatment with voretigene neparvovec

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Purpose: To report clinical outcomes of gene therapy with voretigene neparvovec (VN) in a large cohort of patients with inherited retinal dystrophy caused by biallelic *RPE65* mutations.
Methods: Analysis of clinical records, ancillary testing, and multimodal retinal imaging of patients treated with voretigene neparvovec (VN) in a multi-center, prospective, observational study. Efficacy was evaluated using outcome measures including scotopic full-field stimulus threshold testing (FST), best-corrected visual acuity (BCVA), low-luminance visual acuity (LLVA), semi-automated kinetic perimetry (SKP), and microperimetry.
Results: Twenty-three patients (n=36 eyes) treated between 2021 and 2023 were included for analysis, of whom 19 patients (n=31 eyes) completed the 24-month follow-up. Postoperative retinal sensitivity, measured with FST, was improved by 1 month of treatment (2 log cd.s/m^2). BCVA did not show improvement, while LLVA demonstrated significant gains (0.41 LogMAR). SKP was highly variable without a clear trend. Mean retinal sensitivity, assessed using microperimetry, was consistently measurable in 6 eyes and showed improvement at 24 months (3.24 dB). Progressive chorioretinal atrophy observed in almost all eyes has not adversely affected visual outcomes following VN treatment for at least 24 months.
Conclusion: Data on VN in a clinical setting show sufficient effectiveness of VN with persistence of improvements in retinal sensitivity and LLVA up to 24 months. The long-term effect of progressive chorioretinal atrophy on visual function will require ongoing monitoring.

Evaluating cone density and photoreceptor outer segment length in Stargardt disease

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Introduction: Adaptive optics flood illumination ophthalmoscopy (AO-FIO) and optical coherence tomography (OCT) facilitate detailed examination of photoreceptor structure. They may be useful for developing biomarkers to assess the efficacy of emerging treatments for Stargardt disease (STGD1). This study was performed to investigate the correlation between cone density measured on AO-FIO images and photoreceptor outer segment (POS) length measured on OCT images, and to compare these measurements between STGD1 patients and healthy controls.
Patients and methods: A cross-sectional study was conducted on STGD1 patients and healthy controls. Cone density and POS length were measured at predefined locations on AO-FIO and OCT respectively. Correlation analysis was performed to evaluate relationships between these metrics.
Results: A total of 10 cases with STGD1 and 10 healthy controls were analysed. STGD1 patients exhibited significantly lower cone density and POS length compared to healthy controls. However, no significant correlations were observed between cone density and POS length across all tested retinal locations.
Conclusion: This study found no correlation between cone density measured on AO-FIO images and POS length measured on OCT images. Both cone density and POS length capture distinct photoreceptor characteristics and may offer potential independent biomarkers for progression in STGD1.

The decline in retinal sensitivity occurs at the same rate in both eyes of patients with *USH2A*-related retinitis pigmentosa

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Purpose: Mesopic microperimetry (mMP) measurement of retinal sensitivity (RS) shows promise as potential endpoint for clinical trials in Usher syndrome type 2A-related retinitis pigmentosa (*USH2A-RP)*. Locally administered (gene-)therapies targeting *USH2A-RP* are upcoming, paving the way for paired-eye trial designs using inter-eye correlations to enhance the efficiency of the trial design. This study explores these inter-eye correlations and RS decline in *USH2A*-RP.
Methods: The Characterizing Rate of progression in USHer syndrome (CRUSH) study included 37 participants (71 eyes) with *USH2A*-RP. After exclusions, inter-eye symmetry was analysed in 31 participants (62 eyes) using mixed models. RS was defined as rate of change in total mean sensitivity, in the central 10-degree subfield, and baseline functional transition points (FTPs).
Results: The intraclass correlation coefficient (ICC) of the mean RS decline rate was 0.87 (−0.353 dB/year); mean inter-eye difference was −0.105 dB/year (95% CI -0.220 – 0.010 dB/year).
The RS decline rate within the central 10 degrees showed an ICC of 0.85 (−0.361 dB/year); the mean inter-eye difference was −0.013 dB/year (95% CI -0.154 – 0.127 dB/year).
Within the FTPs, the ICC measured was 0.80 (−1.203 dB/year); the mean inter-eye difference was 0.113 dB/year (95% CI -0.137 –0.363 dB/year).
Conclusions: The ICC between eyes was deemed good (ranging from 0.75 to 0.90) across all methods, with mean differences consistently near zero and 95% confidence intervals including zero. These findings indicate that the rate of RS decline is similar between both eyes in patients with *USH2A*-RP, supporting the potential use of a paired-eye-design in future clinical trials.

Adaptive optics retinal imaging in retinitis pigmentosa patients

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Purpose: To evaluate the cone photoreceptor mosaic using adaptive optics (AO) in retinitis pigmentosa (RP) patients and correlate findings with OCT.
Patients and Methods: 20 RP patients (n=37 eyes) with 9 different causal genes were included in a prospective cohort study. Images were carried out with the rtx1 AO retinal camera (Imagine Eyes). Four rows of images were acquired at 3° superior (S), 1° S, 1° inferior (I), and 3° I, eccentricity. For the rows at 3° S/I, images were taken at 2° nasal (N), 0°, and 2° temporal (T). For the rows at 1°

S/I, images were acquired from 2° N to 10° T in 2° intervals. Outcomes included specific RP cell morphology and AO patterns correlated with OCT.
Results: The median age was 41 years (IQR 30–54), the median best-corrected visual acuity was 0.8 (IQR 0.5–0.9) decimal. Cone photoreceptors and -spacing were abnormal and irregular in all eyes. Hyporeflective cones were present in 20 eyes. On OCT, disruptions in the ellipsoid zone (EZ), interdigitation zone (IZ), and retinal pigment epithelium (RPE) were noted. Homogeneous atrophic patterns were seen in 26 eyes and could be correlated with an absent EZ, IZ and disturbed RPE on OCT.
Conclusions: Cone photoreceptor mosaic is severely disturbed in RP patients, independent of genetic cause. Hyporeflective cones may indicate cells in apoptosis. Typical homogeneous patterns were present in 70% of cases. These findings could be correlated with OCT.

Test–retest variability of the full-field stimulus test in patients with retinitis pigmentosa

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Purpose: To evaluate test–retest variability (TRV) of the full-field stimulus test (FST) in patients with retinitis pigmentosa (RP) and poor best-corrected visual acuity (BCVA; ≤20/50 Snellen; ≥0.40 logMAR), and to assess the reliability of FST as a clinical endpoint in future RP trials.
Methods: In this prospective cohort study, 28 patients with clinically diagnosed RP performed FST on two different days, two weeks apart. Retinal sensitivity thresholds on FST for blue, red, and white stimuli were determined, and the difference between blue and red thresholds was calculated to identify photoreceptor mediation type. TRV was analysed using Bland–Altman analyses and Coefficients of Repeatability (CoRs).
Results: Patients had a median age of 38.0 years (interquartile range [IQR]: 24.5–57.5), and a median BCVA of 1.1 logMAR (IQR: 0.6–1.2). Following Bland–Altman analyses, biases across the FST measurements were minimal, ranging from −0.02 to 0.02 log units. The CoRs were ± 0.38 log units (95% confidence interval [CI]: 0.30–0.52) for blue stimuli, ±0.30 log units (95%CI: 0.23–0.40) for red stimuli, ±0.59 log units (95%CI: 0.47–0.81) for white stimuli, and ± 0.30 log units (95%CI: 0.24–0.41) for the blue-red difference. Patients with mixed photoreceptor mediation had a higher TRV than those with rod or cone mediation. Age was a significant predictor of TRV on blue light FST ($p = 0.028$).
Conclusions: These findings support the use of FST as a reliable measurement of retinal sensitivity in patients with RP, and provide a strong foundation for developing clinical endpoints for FST based on the CoRs to improve assessment of changes in RP, for example before and after treatment.

REPEAT Study: Relationship between the full-field stimulus test and self-reported visual function in patients with retinitis pigmentosa

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Purpose: To determine the relationship between the full-field stimulus test (FST) and self- reported visual function using the Michigan Retinal Degeneration Questionnaire (MRDQ) in patients with retinitis pigmentosa (RP).
Methods: In this cross-sectional study, patients with clinically diagnosed RP ($n = 31$) performed FST to determine retinal sensitivity thresholds for blue, red, and white stimuli. The difference between the blue and red thresholds was used to identify photoreceptor mediation type. Patients completed the MRDQ from which disability (Θ) scores were derived across seven visual function domains. Correlations between the FST thresholds and MRDQ domain Θ-scores were analysed using Spearman's rank correlation.
Results: The median age was 38.0 years, and photoreceptor mediation was rod-based in 11 patients (35.5%), cone-based in 7 patients (22.6%), and mixed in 13 patients (41.9%). The highest disability scores were reported in the domains of '*mesopic peripheral function*' and '*scotopic function*'. Significant correlations were found between all chromatic stimuli thresholds and the MRDQ domains of '*scotopic function*', '*mesopic peripheral function*', and '*photopic peripheral function*'. The strongest correlations of these domains were observed with the blue FST ($p < 0.001$). The threshold on blue stimulus FST and age were significant predictors of the domain scores on '*scotopic function*' ($p < 0.001$), '*mesopic peripheral function*' ($p < 0.001$), and '*photopic peripheral function*' ($p < 0.001$).
Conclusions: Strong correlations between MRDQ domains related to rod function and FST were found in patients with RP. These findings confirm that FST can be used as an informative and clinically relevant endpoint in RP trials when evaluating therapeutic interventions.

Phacoemulsification in patients with end-stage glaucoma

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Purpose: To evaluate whether phacoemulsification can be safely performed in patients with end-stage glaucoma.
Patients: Ten patients with end-stage glaucoma and cataract underwent cataract surgery.

Methods: Inclusion criteria included patients with end-stage glaucoma and cataract. Phacoemulsification was performed using the Centurion® (Alcon Laboratories, USA) device with the Active Sentry® handpiece, and a maximum infusion pressure of 22 mmHg was maintained throughout the procedure.
Results: The average preoperative visual acuity was 0.29 (Snellen linear), while the postoperative average was 0.49, resulting in an average improvement of 0.21. The 95% confidence interval for this improvement was [0.14, 0.45]. All patients retained their central visual field after surgery.
Conclusions: Phacoemulsification with a low infusion pressure appears to be a safe and effective method for treating cataract in patients with end-stage glaucoma.

Carotid artery disease increases the risk for open-angle glaucoma

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Purpose: Studies suggest that open-angle glaucoma (OAG) patients have altered hemodynamics of the carotid and retinal arteries. We aimed to investigate the effect of carotid artery stenosis on incident OAG.
Methods: We analysed data from the Rotterdam Study, a prospective population-based cohort study. The carotid arteries were assessed by measuring the carotid intima-media thickness (cIMT). Using multivariable Cox proportional hazard models, we estimated the effect of the standardized cIMT on the risk to develop OAG. We additionally estimated the association between the cIMT at baseline and intraocular pressure (IOP), macular retinal nerve fibre layer (mRNFL), and macular ganglion cell complex (mGCC) thickness at end of follow-up using linear mixed models.
Results: In total, 9459 non-OAG controls were included and 193 incident OAG cases were detected during 89 665 person-years of follow-up. Average age ± standard deviation at baseline was 62.1 ± 7.9 years and 66.5 ± 7.7 years, respectively. Higher cIMT significantly increased the risk for OAG, with a Hazard Ratio (95% confidence interval) of 1.17 (1.00–1.36) per standard deviation increase in cIMT. Additionally, the mRNFL thickness was 0.17 (0.01–0.34) μm thinner with each standard deviation increase in cIMT. Within OAG cases, cIMT was associated with a younger age at diagnosis (beta [95% CI] = −1.01 [−1.84 to −0.17]; $p=0.018$). There was no statistically significant association between cIMT and IOP or mGCC thickness.
Conclusions: Our study demonstrates that a higher cIMT increases the risk for OAG. In patients with OAG and unexplained continuous progression, a screening ultrasound exam of the carotid arteries could be considered.

Outcomes of combined phacoemulsification with excimer laser trabeculostomy in patients with glaucoma

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Introduction: To investigate the effect of Excimer Laser Trabeculostomy (ELT) as a surgical treatment for patients with glaucoma.
Patients and Methods: Patients with both cataract and glaucoma underwent combined phacoemulsification with lens implantation and Excimer Laser Trabeculostomy (ELT, Elios Vision Inc., Los Angeles, CA, USA). Data was included for analysis if a minimum follow-up period of 3 months was available.
Results: A total of 75 eyes from 44 patients were treated. The mean preoperative intraocular pressure (IOP) was 16.8 ± 5.3 mmHg. Postoperative IOP was reduced to 16.5 ± 7.4 mmHg at 1 week, 13.2 ± 3.5 mmHg at 1 month, 12.7 ± 1.4 mmHg at 3 months, and 12.2 ± 2.3 mmHg at 6 months. The mean number of glaucoma medications decreased from 2.3 ± 1.2 preoperatively to 1.8 ± 1.3 at 1 week, 1.7 ± 1.2 at 1 month, 1.4 ± 1.1 at 3 months, and 1.3 ± 1.3 at 6 months. Both IOP and the number of medications were significantly lower at all time points from 1 month postoperatively compared to preoperative levels ($p<0.001$). Reversible microhyphema was observed in 2 eyes, and transient IOP spikes occurred in 8 eyes. No other complications were noted during the follow-up period.
Conclusion: In this patient cohort, Excimer Laser Trabeculostomy proved to be a safe treatment option for patients with cataracts and glaucoma. Further research and long-term follow-up are needed to evaluate the long-term effects of this treatment.

Five-year outcomes of the PreserFlo MicroShunt in patients with open-angle glaucoma

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Purpose: To evaluate the five-year outcomes of the PreserFlo MicroShunt (MicroShunt) as a surgical intervention for open-angle glaucoma.
Patients and Methods: A retrospective case series was conducted. Consecutive patients who underwent a stand-alone MicroShunt implantation at the University Eye Clinic of Maastricht were included. If the procedure was

performed in both eyes, only the first operated eye was included for analyses. MicroShunt implantations were augmented with 0.2 mg/mL mitomycin-C. The primary outcome measure was intraocular pressure (IOP) during follow-up. Secondary outcomes included the use of IOP-lowering medications, surgical success rates, reoperation rates, and postoperative complications.
Results: A total of 66 eyes were included. Patients had primary open-angle glaucoma (88%) and pigmentary glaucoma (12%). Most patients had moderate to advanced glaucoma based on visual field mean deviation. Mean IOP (95% confidence interval) significantly decreased from 21.8 mmHg (20.8–22.8) at baseline to 13.2 mmHg (11.8–14.6) at five years ($p<0.001$). The mean number of IOP-lowering medications reduced from 2.5 (2.1–2.9) at baseline to 0.9 (0.5–1.2), 1.0 (0.7–1.4), and 1.1 (0.7–1.5) at three, four, and five years, respectively (all $p<0.001$). Twelve eyes (18%) required needling or surgical revision. Further IOP-lowering surgery was necessary in 19 eyes (29%). Postoperative complications, including early hypotony, shallow anterior chamber, and hyphema, were generally mild and self-limiting.
Conclusions: The MicroShunt demonstrated sustained efficacy in reducing IOP and the use of IOP-lowering medications over five years, with a good safety profile. However, nearly one-third of eyes required additional surgical interventions.

The role of dutch healthy diet in glaucoma prevalence: Insights from the maastricht study

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Purpose: To evaluate the association between adherence to the Dutch Healthy Diet Index (DHD-index) and glaucoma prevalence. The study also explored whether specific dietary components, including lutein intake, are associated with glaucoma prevalence.
Methods: This cross-sectional case–control study was performed within the population-based Maastricht Study cohort. Dietary data was collected using a validated food frequency questionnaire, and the DHD-index score was assessed by evaluating adherence to 15 dietary components. Glaucoma status was determined via visual field tests and expert consensus. Logistic regression models were used to estimate the association between DHD scores, individual dietary components, and glaucoma prevalence, adjusted for co-variables of age, sex, intraocular pressure (IOP), waist circumference, and other lifestyle factors.
Results: A total of 6831 participants (aged 59.4 ± 8.8 years; 50.1% female) were included in the analysis, of which 736 (10.8%) were identified as glaucoma cases. Each 10-point increase in the DHD-index score was associated with a 12.5% lower odds of glaucoma prevalence (OR [95% CI]: 0.875 [0.827–0.927]). Among individual dietary components, higher intake of vegetables, wholegrain products, nuts, fish, and tea, were significantly associated with reduced glaucoma prevalence, whereas high sugar and processed meat intake tended to have increased odds of glaucoma. No significant association was found between total lutein intake and glaucoma prevalence, but the lutein intake from fruit and nuts showed significant differences in glaucoma and control group.
Conclusions: Adherence to the DHD-index, reflecting an overall high-quality diet, was inversely associated with glaucoma prevalence. These findings underscore the importance of studying dietary patterns in glaucoma research and suggest that a combination of healthy foods is associated with a lower odd of glaucoma.

Generating pluripotent stem cell-derived retinal ganglion cells via retinal organoids

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Purpose: To develop a robust protocol to generate patient-derived retinal ganglion cells (RGCs) for in vitro modelling and regenerative optic nerve transplantation.
Methods: Retinal organoids (ROs) were generated from H9 cells in the presence or absence of 50 ng/mL BDNF from week (wk) 6 to 15. At wk. 7, 10, and 15, RGCs were isolated by magnetic Thy1 immuno-affinity isolation and cultured for 14 days in Brainphys or RGC media.
Results: mRNA expression of early retinal markers PAX6, SIX3, and VSX2 was ≥500-fold enriched in ROs compared to stem cells and stable throughout wk. 7–15, while RGC markers ATOH7, ISL1, BRN3B, and RBPMS peaked at wk. 7 and decreased until wk. 15. Immunostained area for Islet1, BRN3A, and RBPMS in RO cryosections was highest in wk. 7 (23, 18, and 8% of DAPI-positive signal) and decreased to ≤3% by wk. 13. Thy1 immunostaining was found in the ganglion cell but also the photoreceptor layer. $Thy1^+$ cell yield increased ca. 3-fold from wk. 7 to 10 or 15. Higher cell percentages of the $Thy1^+$ fraction stained positive for ISLET1, BRN3A, and RBPMS at wk. 7 and 10 than 15. BDNF supplementation had no effect on RGC yield. RGC survival was similar in Brainphys and RGC media, but growth of cells with non-neuronal morphology was observed in RGC media.
Conclusions: Thy1 immunoaffinity isolation enriches but does not purify RGCs. Wk 7–10 is an ideal window for RGC isolation with highest yield at wk. 10. Brainphys media may be preferred for RGC maintenance.

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Mitochondrial DNA analysis in glaucoma: association of D-loop variants with a primary open-angle glaucoma subgroup

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Purpose: Primary open-angle glaucoma (POAG) is an optic neuropathy caused by degeneration of the optic nerve-forming neurons, the retinal ganglion cells (RGCs). High IOP and aging have been identified as major risk factors; yet the POAG pathophysiology is not fully understood. Since RGCs have high energy requirements, mitochondrial dysfunction may put the survivability of RGCs at risk. We explored in blood DNA whether mtDNA variants and their distribution throughout the mtDNA regions/genes could be risk factors for POAG.

Methods: The mtDNA was sequenced from age- and sex-matched study groups, being high-tension glaucoma patients (HTG, $n=71$), normal-tension glaucoma patients (NTG, $n=33$), ocular hypertensive subjects (OH, $n=7$), and cataract controls (without glaucoma; $n=30$), all without remarkable comorbidities.

Results: No association was found between the number of mtDNA variants in genes encoding proteins, tRNAs, rRNAs, and in non-coding regions between the different study groups. Next, variants that controls shared with the other groups were discarded. A significantly higher number of exclusive variants was observed in the non-coding *D-loop* region for the HTG group (~1.23 variants/subject), in contrast to controls (~0.35 variants/subject). The most frequent variants exclusively found in glaucoma patients were m.72T>C, m.16163A>G, m.16186C>T, m.16298T>C and m.16390G>A. The *D-loop* plays a crucial role in mtDNA replication and transcription, and variants in this region have been suggested to affect these processes.

Conclusions: In conclusion, we found that variants in the *D-loop* region may be a risk factor in a subgroup of POAG, which could possibly affect mtDNA replication.

A metabolome-wide study of open-angle glaucoma reveals upregulated lipid metabolism pathways

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Purpose: Metabolomics enables the discovery of biomarkers and mechanisms underlying diseases. We investigated metabolic alterations associated with open-angle glaucoma (OAG) to identify potential therapeutic targets.

Methods: Data were derived from two cohorts of the Rotterdam Study ($N=2559$; mean±standard deviation age 80±9 years; 57.3% female), including 65 (2.5%) OAG cases. Intraocular pressure (IOP) was adjusted for IOP-lowering treatment, and retinal layer thicknesses were quantified via OCT using an in-house algorithm (EyeNED). Genetic risk scores (GRS) for OAG and IOP were based on meta-GWAS data. Plasma levels of 940 metabolites were measured from fasting blood samples and metabolite set enrichment analysis was used to identify key metabolic pathways. Associations with pathways and individual metabolites were assessed using logistic and linear regression, adjusted for age, sex, BMI, and sub-cohort.

Results: The lipid super pathway (LSP) was enriched in participants with a higher IOP (normalized enrichment score [NES]: 2.34), a thinner retinal nerve fibre layer (RNFL; NES: −1.41), a higher OAG GRS (NES: 1.39), and a higher IOP GRS (NES: 1.24). Within the LSP, androgenic steroids, endocannabinoids, dicarboxylate, long-chain monounsaturated fatty acids (FAs), long-chain polyunsaturated FAs, long-chain saturated FAs, medium-chain FAs, phosphatidylethanolamine, pregnenolone steroids, and sphingomyelins were associated with OAG-related parameters. A higher IOP GRS was associated with the upregulation of 14 LSP-metabolites, five of which mediated the GRS-IOP relationship with 1.2–1.7% (95%CI: 0.1–3.8%).

Conclusions: Lipid metabolic pathways were found to be significantly associated with OAG-related parameters, highlighting the potential of metabolites as potential biomarkers and therapeutic targets.

Nitric oxide bioavailability and excitotoxicity in glaucoma and tinnitus: A case–control study

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Purpose: To determine if low nitric oxide bioavailability and excitotoxicity may be common mechanisms behind glaucoma and tinnitus.
Methods: Nitric oxide itself diffuses quickly and is difficult to quantify, however, certain metabolites can be measured as biomarkers of nitric oxide production. We assessed serum levels of L-arginine, L-ornithine, L-citrulline, L-glutamate, and L-glutamine in 23 patients with primary open angle glaucoma (14 with tinnitus and 9 without) and 20 controls (13 with tinnitus and 7 without). The groups were age-similar. A quantile regression was performed for each biomarker with presence of glaucoma and presence of tinnitus as independent variables.
Results: L-arginine was higher in the glaucoma group (68.0 [59.7 to 76.2] μmol/L) compared to the controls (54.5 [35.7 to 73.2] μmol/L), but this difference was not significant (p=0.14). There were no significant differences in the other nitric oxide biomarker concentrations between glaucoma and controls: L-ornithine (p=0.78), L-citrulline (p=0.77), L-glutamate (p=1.0), and L-glutamine (p=0.52). Presence of tinnitus also did not significantly affect the biomarkers.
Conclusions: This research did not show abnormal levels of nitric oxide biomarkers in glaucoma and tinnitus patients. However, there was a trend for increased L-arginine in glaucoma patients compared to controls.

Dietary intake of vitamin B3 (niacin) and glaucoma incidence: The Rotterdam Study

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Purpose: There is increasing evidence for a role of vitamin B3 (niacin) in the pathophysiology of neurodegenerative diseases, including glaucoma. We aim to evaluate the association between dietary niacin intake and the incidence of open-angle glaucoma (OAG).
Methods: We used data from the prospective, population-based Rotterdam Study. Participants were followed for incident OAG since 1991, with intervals of 4–5 years. Participants included (N=6742) were free of glaucoma at baseline, had at least one ophthalmic follow-up examination, and had complete data on dietary intake (food frequency questionnaires). A total of 162 participants developed OAG during follow-up. Niacin intake was adjusted according to the nutrient residual method. Associations of niacin with OAG and intraocular pressure (IOP) were assessed using logistic and linear regression analyses. All analyses were adjusted for age, sex, BMI, diet quality, caloric intake, and follow-up time.
Results: Each milligram increase of niacin was associated with a 6.2% reduction in OAG incidence (Odds ratio [OR] with 95% confidence interval [CI]: 0.94 [0.90–0.98]). There was a significant trend between higher intake of niacin and lower OAG incidence (p-trend=0.02). Especially participants with the highest intake (Q5: mean 23.3 mg/day) had a lower incidence (OR [95%CI]: 0.43 [0.21–0.86]) compared to participants with the lowest intake (Q1: mean 10.0 mg/day). There was no significant association between niacin intake and IOP.
Conclusions: Dietary niacin intake is inversely associated with OAG incidence. We did not observe an association between niacin and IOP, which supports existing literature and suggests that the protective effect of niacin may be IOP-independent.

Efficacy and tolerability of topically administered Latanoprost/Netarsudil (Roclanda) in tertiary glaucoma center

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Purpose: To assess the efficacy and tolerability of topically administered latanoprost/netarsudil (Roclanda) of glaucoma patients in a tertiary care center.
Methods: This retrospective cohort study included glaucoma patients treated with Roclanda from June 2024 to February 2025 with at least 6 weeks follow-up. Clinical data were collected at baseline and follow-up visits through 6 months. If the drops were not tolerated, they were stopped immediately after visit. Possible side effects were recorded at each visit. The effectiveness of roclanda was assessed by baseline pressure measurement relative to 6 weeks later.
Results: Sixty patients treated with Roclanda have been observed. Of these, 43 patients were receiving treatment with maximal glaucoma therapy and 17 patients with 2 different drops. Mean age was 66 ± 15 years and intraocular pressure (IOP) 22 ± 15 mmHg.
Of all patients, 4 stopped taking Roclanda because they developed complaints of redness combined with irritation of the eye. If only redness was present, it was discussed with the patient how many resulting problems he or she had. The average pressure drop of Roclanda was 2 mmHg 6 weeks after started.
Conclusions: Roclanda is an effective additive to glaucoma drops in a tertiary care center and in few cases produces irritation symptoms leading to discontinuation of the drops.

Cytokine profiling in glaucoma filtration surgery fibrosis associated failure

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Introduction: Glaucoma filtration surgery (GFS) is an effective treatment for glaucoma, however, fibrosis-induced bleb failure remains a challenge. Understanding the role of biomarkers in fibrosis is critical for improving surgical outcomes. In a systematic review of biomarkers in aqueous humour (AH), tear fluid (TF) and Tenon's (TN) and their association with fibrosis and GFS outcome, we identified 20 studies and 12 biomarkers - MCP-1, TGF-β, VEGF, IFN-γ, MMP-3, SPARC, IL-1β, IL-2, IL-6, IL-8, IL-10, and IL-18 - which were consistently significantly elevated in failed GFS, except for IFN-γ, which was the only biomarker significantly decreased.
Methods: Cytokine profiling of AH was conducted on 46 trabeculectomy (TE) patients and 14 cataract controls. In TE patients, biomarker levels were compared between successful (n=36) and failed (n=10) outcomes at 3, 6, and 12 months post-surgery.
Results: TE patients exhibited significantly elevated MMP-3 ($p<0.001$), TGF-β2 ($p=0.036$), MCP-1 ($p=0.002$), IL-8 ($p<0.001$), and IL-18 ($p=0.002$) compared to controls. Among TE failures, IL-1β ($p<0.001$), MCP-1 ($p=0.012$), and IL-10 ($p=0.032$) were significantly elevated at 3 months, while IL-1β ($p<0.002$), MCP-1 ($p=0.004$), and IL-8 ($p=0.016$) were significantly elevated at 6 months. At 12 months, IL-1β ($p<0.011$) and IL-18 ($p=0.029$) were significantly elevated.
Conclusion: IL-1β was consistently associated with TE failure up till 1 year post-surgery. Further studies are needed to validate these biomarkers and develop targeted therapies to improve GFS outcomes.

In vitro studies to explore the potential of rescuing dying RGCs

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Purpose: Glaucoma is a neurodegenerative disease where various triggers initiate cascades of secondary events, ultimately causing retinal ganglion cell (RGC) death. Apoptosis is a major mechanism of RGC death, traditionally considered irreversible. However, recent studies suggest that apoptosis can be reversed under certain conditions. This study investigates whether dying RGCs can be rescued, focusing on the point of no return in apoptosis and mechanisms of recovery.
Methods: In vitro apoptotic models were established using ethanol to induce cell death in RGCs and other neuronal cell lines. Live cell imaging and fluorescent probes (Mito-Tracker, TMRM, DCFDA, Fluo-8 AM) were used to observe apoptotic injuries. During ongoing apoptosis, ethanol was washed out, and cells were cultured in fresh medium to assess recovery potential. Live cell microscopy monitored the entire death and recovery process.
Results: Neuronal cells, including RGCs, exhibited an intrinsic capacity to recover from apoptotic injuries prior to mitochondrial cytochrome c release. Injuries such as mitochondrial fragmentation, membrane potential loss, and elevated intracellular Ca2+ were reversible. However, cytochrome c release marked the point of no return, beyond which recovery was not observed. Mitochondrial biogenesis was critical for recovery, as inhibiting this process significantly reduced cell survival and neurite regrowth.
Conclusions: Dying RGCs retain intrinsic recovery potential if key apoptotic events, like cytochrome c release, are prevented. Mitochondrial dynamics, particularly biogenesis, play a pivotal role in cell recovery. These findings highlight the possibility of targeting dying RGCs to develop therapies for preserving vision in glaucoma.

Deep learning analysis of retinal vasculature from colour fundus photography reveals diagnostic and predictive value for open-angle glaucoma

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Purpose: Previous studies have reported vascular changes in open-angle glaucoma (OAG) patients. We investigated the key vascular features most relevant for glaucoma and their diagnostic and predictive value.
Methods: We analysed data from the Rotterdam Study, a large prospective population-based cohort study. Vascular features were extracted from colour fundus images using our fully automated deep-learning vascular analysis pipeline (VascX). We analysed the association of the standardized vascular features with OAG using multivariable logistic models. Diagnostic accuracy was assessed through the area under the receiver operator curve (AUC). Using Cox proportional hazard models, we estimated the relative risk of OAG. Prediction of OAG was tested with Harrell's C-statistic.

Results: In total, 12 734 non-OAG controls and 356 OAG cases were included, with an average ± SD age of 65.1 ± 9.8 and 68.5 ± 8.6 years and an average ± SD follow-up time of 10.5 ± 5.9 and 12.7 ± 6.3 years, respectively. Overall, features associated with OAG were arterial, not venous. The AUC (95% CI) for arterial vessel density and arterial central retinal equivalent with OAG were 0.66 (0.63, 0.68) and 0.64 (0.61, 0.66), respectively. The C-statistic (95% CI) for baseline arterial vessel density and OAG incidence was 0.67 (0.63, 0.71); including age and sex into the model increased the concordance (95% CI) to 0.74 (0.72, 0.76).
Conclusions: Our study reinforces the evidence of an association between retinal vasculature and glaucoma and suggests that retinal arteries have diagnostic and prognostic value. Future studies should explore clinical applications of retinal vasculature measurements in OAG management.

Effects of Prolene stent removal timing and length on intraocular pressure in Paul Glaucoma Implant surgery

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Introduction: The use of a temporary 6–0 Prolene intraluminal stent in Paul Glaucoma Implant (PGI) surgery has become a widely adopted technique to prevent early postoperative hypotony. This study examines the effects of stent removal timing and its length on intraocular pressure (IOP).
Methods: This retrospective study analysed 130 PGI surgeries at The Rotterdam Eye Hospital (December 2022 and July 2024). IOP and glaucoma medication use were recorded pre-operatively, before stent removal, at the first follow-up after removal, and six months post-operatively. A linear mixed-effects model assessed the effect of stent removal timing on IOP, adjusting for age, sex, medication use, and time since surgery. A generalized-least-squares-regression model assessed the relationship between stent length and IOP change before and after removal.
Results: The mean (SD) age was 67.6 (12.2) years and 50.8% of the participants were male. The median (IQR) timing of stent removal was 47 (34–59) days post-surgery. The median (IQR) stent length was 6 mm (5–6). All four hypotony cases had stent removal before 6 weeks after PGI-surgery. The duration between PGI surgery and stent removal was not significantly associated with changes in IOP ($p = 0.61$). However, stent length showed a significant association with the change in IOP before and after stent removal ($\beta = 0.96$, $p < 0.001$).
Conclusion: Timing of Prolene stent removal did not significantly affect IOP, but early removal may increase hypotony risk. Surgeons should consider stent length, as longer stent lengths were associated with an approximate 1 mmHg greater reduction in IOP post-removal.

Effect of time at room temperature on complement activity in aqueous humour samples

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Purpose: The complement system, part of the immune system, is activated through cascades of temperature-sensitive proteins. This study aims to investigate the effect of room temperature (RT) storage time on complement activity in aqueous humour (AH) samples representing local ocular condition, estimate this potential systematic error impacting future research on complement-related mechanisms in glaucoma progression, and explore control methods.
Methods: Each AH sample from 16 cataract surgery eyes was divided into 5 aliquots and treated differently before freezing: <2-minute RT incubation, 15-minute RT incubation, 30-minute RT incubation, 30-minute 37°C incubation, and 30-minute RT incubation with EDTA. The concentrations of 15 complement proteins (measured by multiplex ELISA) and the concentration ratios of 3 proteolytic products to their precursor proteins were pairwise compared between the other aliquots and the <2-minute RT incubation aliquot of the same sample.
Results: Statistical significance was defined as $p < 0.05$. In AH with 15-minute RT incubation, most complement proteins concentrations showed no changes except for mild increases in C3, C5, C5a, and a mild decrease in the concentration ratio C3b(iC3b)/C3. In AH with 30-minute RT incubation, most proteins increased, and more ratios changed. The above changes were mild (with fold changes all below 1.3) and non-stepwise over time. With the same 30-minute incubation, adding EDTA in advance prevented the increases in protein concentrations that occurred without it.
Conclusions: RT time mildly affects complement activity in AH samples but does not impact studies with intergroup differences greater than 2-fold. EDTA can inhibit this effect.

Delayed visual maturation: Understanding the causes and visual prognosis

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Introduction: Delayed visual maturation (DVM) is a rare condition where infants initially appear blind but later achieve visual milestones, typically resolving by 4–6 months without intervention. Causes include prematurity, developmental delay, and neurological or ocular abnormalities (e.g., congenital nystagmus, albinism). In

some cases, it occurs in isolation with normal neurological and ocular findings. Advances in diagnostic tools, such as handheld optical coherence tomography (HH-OCT) and handheld electroretinography (HH-ERG), allow earlier differentiation of DVM causes. This study explores the aetiologies and visual prognosis of DVM.
Methods: This retrospective study was conducted at the Bartiméus Diagnostic Centre for Complex Visual Disorders in Zeist, Netherlands. Data from the records of 130 children (ages 0–3 years) diagnosed with DVM were analysed. Visual acuity was assessed in relation to age and underlying causes.
Results: Albinism was the most common cause of DVM, with affected infants typically improving between 4 and 6 months. Children with DVM linked to developmental delay, systemic conditions, or neurological abnormalities showed significant variability in recovery and outcomes.
Conclusion: This study highlights the diverse causes of DVM and emphasizes the importance of thorough diagnostic evaluation. In the absence of ophthalmological or neurological abnormalities, prognosis is generally favourable. However, developmental or systemic conditions result in more uncertain outcomes, requiring individualized management strategies.

The association between red blood cell transfusion timing and the development of retinopathy of prematurity: Application of the two-phase theory

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Purpose: Red blood cell (RBC) transfusions are believed to be associated with retinopathy of prematurity (ROP), but the underlying mechanism is not fully understood, partly due to lack of information on RBC transfusion timing. The aim was to determine the association between timing and amount of RBC transfusions and development and severity of ROP.
Methods: This retrospective dual-center cohort study included 1177 neonates born in 2004–2022 with a gestational age at birth ≤28 weeks. Primary outcomes were any stage ROP and severe ROP based on maximum staging. Phase I of ROP was defined as ≤32.0 weeks postmenstrual age and phase II as >32.0 weeks postmenstrual age. Logistic regression analyses were adjusted for gestational age at birth, small for gestational age, mechanical ventilation duration, postnatal corticosteroids, sepsis and necrotizing enterocolitis.
Results: Multivariate analysis showed independent associations with severe ROP for RBC transfusion (OR 5.1; 95% CI 1.8–14.3), number of RBC transfusions (OR 1.2; 95% CI 1.1–1.3), RBC transfusion in phase I (OR 2.5; 95% CI 1.2–5.3), number of RBC transfusions in phase I (OR 1.2; 95% CI 1.0–1.3), RBC transfusion in phase II (OR 4.1; 95% CI 2.7–6.3) and number of RBC transfusions in phase II (OR 1.9; 95% CI 1.5–2.4).
Conclusion: Based on maximum ROP staging, RBC transfusions in phases I and II are both associated with a high risk of severe ROP. A randomized controlled trial is urgently needed to determine the potential effect of RBC transfusions in phase II on ROP progression.

Global axial length growth charts: A comprehensive study across continents in the CREAM-KIDS consortium

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Purpose: Myopia is a growing global public health concern due to complications associated with axial elongation. Current prevention and management strategies focus on axial length, yet reference growth patterns remain unclear and vary by region and sex. This study aims to create region- and sex-specific axial length growth charts using data from the global Consortium of Refractive Error and Myopia research in Children (CREAM-KIDS).

Methods: Data from 172 788 children and young adults across 16 cohorts were analysed. Asian cohorts (N=128 309) were analysed separately, while Australian (N=6123) and European cohorts (N=34 183) were grouped as "Western." Axial length was measured using IOL-500, IOL-700, or Lenstar devices. Age ranged from 3 to 18 years in Asians and 5 to 30 years in Western cohorts. Growth curves were constructed for regions and sexes by fitting multi-Gaussian models and generating regional centile curves using weighted cubic splines, smoothed with the Generalized Akaike Information Criterion.

Results: A total of 290 757 axial length measurements were analysed: 251104 from Asians, 6508 from Australians, and 30 998 from Europeans. Axial length increased with age, with Asian males showing faster elongation in early childhood compared to other groups. Females consistently had lower axial lengths than males across all regions.

Conclusions: These comprehensive axial length growth charts reveal significant regional and sex differences. They provide valuable references to distinguish normal from excessive growth, supporting tailored myopia management strategies.

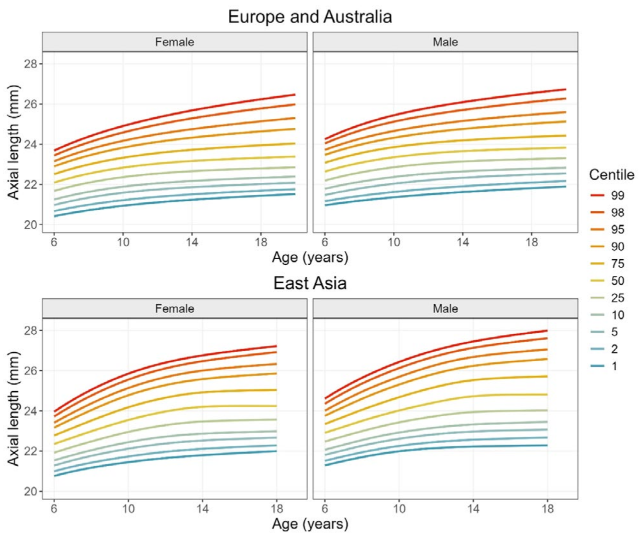

Figure 1. Region and sex specific growth curves for children 6–18 years old.

Safety and efficacy of high dose (0.5%) vs. low dose (0.05%) atropine in European children: interim results of the MAD trial

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Purpose: To evaluate the safety and efficacy of atropine 0.5% vs. 0.05% in European children with progressive myopia after 6 and 12 months.
Methods: In this ongoing randomized trial, 320 children (mean age 8.67 ± 1.48 years, 50.3% boys) were recruited across 20 centers in The Netherlands. All participants received masked atropine treatment and photochromic multifocal spectacles. Safety measures included visual acuity (VA), pupil diameter, intraocular pressure (IOP), and heart rate. Efficacy was assessed by changes in axial length (AL) and spherical equivalent refraction (SER).
Results: VA remained stable (median decimal VA 1.0). Pupil diameter increased significantly (+2.20 ± 1.58 mm at 6 months, +2.13 ± 1.54 mm at 12 months, $p < 0.001$), with minimal changes in IOP and heart rate. Adverse events included photophobia (4.2%) and allergic reactions (2.4%). Mean AL change was +0.025 ± 0.12 mm at 6 months and + 0.126 ± 0.17 mm at 12 months. SER improved by +0.29 ± 0.34 D at 6 months and + 0.055 ± 0.44 D at 12 months.
Conclusions: Atropine 0.5% and 0.05% were well-tolerated with minimal side effects. Both doses effectively reduced axial elongation and myopia progression. These findings support the safe use of higher-dose atropine in European children.

Retinopathy of prematurity screening in The Netherlands: How do we do it?

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Purpose: To gain insight into how retinopathy of prematurity (ROP) screening is performed (with or without camera) and the experience of ophthalmologists in The Netherlands. Secondary, we investigated how ROP screening is incorporated into the ophthalmology training program for residents.
Methods: Separate structured questionnaires were sent to ophthalmologists in 72 hospitals where ROP screening is carried out, and to all ophthalmology residents in The Netherlands.
Results: 36 ophthalmologists from 32 hospitals (44.4%) in The Netherlands responded. 55% of the ophthalmologists perform ROP screening manually. A camera for ROP screening is used in 100% (n = 5) of NICU departments, 70% (n = 7) of post-ICU/high-care departments, and 6% (n = 1) of medium-care departments. The screening workload is highest in NICUs and high-care/post-ICU departments, with the majority (80%) having a camera. There is a high level of certainty reported among ophthalmologists for both manual and camera-based ROP screening. 28 residents from 7 training centers responded: 92.6% (n = 25) have learned ROP screening during their training and were trained with a fundus camera, 25.9% (n = 7) also with indirect fundoscopy, and 33.3% (n = 9) also with binocular indirect fundoscopy. 77.9% expect to be able to perform ROP screening after completing their residency, but more than 2/3 of them would need assistance from colleagues.
Conclusions: In The Netherlands, residents are mainly trained with a camera, but a camera is not available in the majority of hospitals. Therefore, the technique of manually performing ROP screening with indirect fundoscopy remains important.

Anti-VEGF for retinopathy of prematurity: A one-year observational analysis of the treatment process and national collaboration

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Purpose: To evaluate the diagnostic and therapeutic processes for anti-VEGF as primary treatment for severe retinopathy of prematurity (ROP), including nationwide collaboration since the implementation of new ROP-guidelines. This is a sub-analysis of a nationwide observational cohort study on ROP treatment.
Methods: Eligible were infants treated with anti-VEGF for ROP in The Netherlands from October 2023 to October 2024. Ophthalmic data, treatment characteristics, timing of treatment and reactivation, and correspondence between ophthalmologists were collected after parental consent.
Results: Data was available for six out of eight eligible infants. In each case, peer consultation between ROP experts from Leiden, Nijmegen and Rotterdam was sought. Treatment decision was a result of one or more of the following indications: very young post-menstrual age (PMA) at treatment indication (n = 5), aggressive ROP (n = 3), maximum vessel growth in zone I (n = 2) or posterior zone II (n = 4), child being unfit for general anaesthesia (n = 2), and inadequate visualization of the retina (n = 2). Median PMA at primary treatment was 33 weeks and 3 days (IQR = 14 days). All six infants received ranibizumab 0.02 mL (10 mg/mL). N = 1 reached full vascularization 97 days after primary treatment without retreatment. Disease reactivation occurred in N = 5;

median reactivation time was 36 days (IQR = 14 days). *N* = 4 received laser as secondary treatment. N = 1 was reinjected with anti-VEGF: 40 days post-reinjection, a vitrectomy (right eye) and (endo)laser (both eyes) were performed.

Conclusions: In The Netherlands, exposure to indications for anti-VEGF as described in the new ROP-guidelines is low. Peer consultation and collaboration between ROP experts is therefore highly recommended.

Ophthalmological findings in children with a primary brain tumour, the visual function up to two years after diagnosis

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Introduction: Visual impairment can be a progressive and irreversible adverse effect in children with a brain tumour. The prevalence and types of ophthalmological findings in these children are unknown, and there are no standardized international guidelines for screening and follow-up. In addition, it is unclear how vision evolves during and after treatment. This study aims to describe the relationship between visual function up to two years after diagnosis, disease status, and different treatments in children with a primary brain tumour.

Methods: In this prospective cohort study, children (0–18 years) diagnosed with a primary brain tumour (May 2019 and September 2021), were enrolled in four hospitals in The Netherlands. A standardized ophthalmological examination was performed within 4 weeks from brain tumour diagnosis. Follow-up visits were planned at 6, 12, 18 and 24 months. The main endpoints were visual acuity and visual field. Data analysis was performed using descriptive statistics.

Results: Of the 170 included children (median age 8.3 years), 78.8% presented with abnormal ophthalmological findings at diagnosis. In total, 13 patients were binocular visually impaired, and 32 had a visual field defect. At 6, 12, 18, and 24 months, binocular visual impairment was present in 4, 2, 4, and 3 patients. Visual field examination was abnormal in 40, 32, 34, and 32 patients, respectively.

Conclusions: These results confirm a high prevalence of abnormal ophthalmological findings in children at primary brain tumour diagnosis and after treatment. Our study emphasizes the need for standard ophthalmological follow-up in this population.

Analysis of follow-up findings in the Early Glasses Study

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Introduction: The relationship between refractive error at age 1 and the risk of developing amblyopia or accommodative esotropia is unknown. This is determined in the Early Glasses Study by examining healthy children at age 1 and follow them to age 4 when visual acuity is measured.

Patients and Methods: Of 865 children recruited at Children's Healthcare Centres (CHCs), 601 were healthy and had received orthoptic examination followed by cycloplegic retinoscopy at 14.5 ± 1.7 months. Fifty-two exceeded >+3.5D spherical equivalent (SE), >1.5D astigmatism, or >1.5D anisometropia and were randomized into wearing glasses or not. Two out of 865 had amblyopia, 10 strabismus. Randomized children are followed-up biannually to age 4 when visual acuity is measured, in all other children visual acuity is measured at CHCs.

Results: Until now, in 173 visual acuity was measured at age 4, 152 of whom had sufficient visual acuity, 1 anisometropia and astigmatism developed later now amblyopia treated with glasses, 1 partial accommodative esotropia and secondary amblyopia, 3 strabismus, 3 glasses immediately after first signs of esophoria, 1 glasses for high hypermetropia, 1 reduced vision, 1 myopia, 10 insufficient visual acuity and appointment with orthoptist to follow. Of randomized children, 7 had an increasing refractive error between age 1 and 4, with a maximum of 2.25D, in 13 refractive error decreased with a maximum of 3.75D.

Conclusion: Of all examined, 10 out of 865 had strabismus at 14 months, 7 of 173 examined at age 4 were added since, as expected, whereas 2 had amblyopia at 14 months, 2 were added since.

Complication rate and long-term visual outcome after paediatric cataract surgery: A retrospective single centre cohort study

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Purpose: To determine the rate of postoperative complications and long-term visual outcome in children undergoing cataract surgery.

Methods: A retrospective cohort study was performed on children from 0 to 16 years old that underwent cataract surgery between January 2010 and July 2024 at the Amsterdam UMC.

Results: The study included a total of 249 eyes of 172 patients. Unilateral cataract was present in 55.2% and bilateral cataract in 44.8% of the patients. The median age at diagnosis was 2.1 months (range: 0.2–179.7 months)

in the group with post-operative aphakia (n=60 patients) and 45.0 months (range: 0.5–185.0 months) in the group with post-operative pseudophakia (n=112 patients) (p<0.001). The median age at surgery was 3.3 months (range: 1.2–179.8 months) in the aphakia eyes and 65.7 months (range: 10.9–187.5 months) in the pseudophakic eyes (p<0.001). The median follow-up in the aphakic eyes was 55.1 months (range: 1.6–214.1 months) compared to 42.6 months (range: 0.7–130.6 months) in the pseudophakic eyes (p=0.115). The median best-corrected visual acuity (BCVA) was 0.7 logMAR (range: −1.1-2.7 logMAR) in aphakic eyes and 0.131 logMAR (range: −0.17 to 3.0 logMAR) in the pseudophakic eyes (p<0.001). Visual axis opacification (VAO) requiring intervention (n=86 eyes (34.5%)) and glaucoma requiring surgery (n=10 eyes (4%)) of which were all aphakic eyes were the most common registered complications.

Conclusions: In this historic single center cohort pseudophakic eyes resulted in a higher level of BCVA compared to aphakic eyes. VAO requiring intervention was present in one in three eyes. Glaucoma requiring surgery appeared in one in twenty-five eyes, which were all aphakic.

The prevalence of visual impairment in The Netherlands: A meta-analysis

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Purpose: In The Netherlands, we do not know the prevalence of visual impairment. Previous estimates performed fifteen years ago were based on foreign research. The aim was to estimate the current prevalence of visual impairment with data from adults in The Netherlands.

Methods: All included databases contained data on vision (visual acuity, self-reported visual ability, or International Classification of Primary Care (ICPC) codes), had the last measurement after 2010, and were roughly representative for the Dutch general population. Self-reported visual ability was categorized as near vision difficulties; distance vision difficulties; either of these; and both. Population-based prevalences were calculated for each database and pooled using meta-analyses of proportions with random-effects models.

Results: Eight databases including 1 814 194 persons were analysed. Prevalences ranged from 0.2 to 0.5% for visual acuity or ICPC codes, 1.1 to 5.9% for difficulties with near vision, 0.6 to 2.7% for difficulties with distance vision, 1.6 to 6.5% for either of these, and 0.2 to 1.0% for both. The pooled prevalence for difficulties with near vision was 1.58% (95% CI: 0.43–3.41), problems with distance vision 0.91% (95% CI: 0.43–1.58), either near or distance problems 2.05% (95% CI: 0.53–4.46), and both near or distance problems 0.49% (95% CI: 0.31–0.71). In all models, heterogeneity was high (>90%). Subgroup analyses indicated moderating effects by data source.

Conclusions: This study provides more accurate estimates of the prevalence of visual impairment in The Netherlands, allowing more insight into future eye care demands and related guidelines for healthcare policy in this high income country.

Modelling diabetic retinopathy using human iPSC-derived RPE and retinal organoids

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Purpose: This study evaluates the long-term effects of hyperglycemia on human iPSC-derived RPE cells and retinal organoids (RO) to develop in vitro models for diabetic retinopathy (DR).

Methods: Two human iPSC lines from healthy subjects were differentiated into RPE and RO. To simulate diabetic retinopathy conditions, iPSC-RPE cells were exposed to high glucose (HG) concentrations (30 mM, 40 mM, and 50 mM) and HG combined with inflammatory factors TNF-α (1 ng/mL) and IL-6 (1 ng/mL) every other day for 2 and 4 weeks. RO at Day 150 were treated with 40 mM HG combined with TNF-α and IL-6 for 2 and 4 weeks. Gene expression related to inflammation, oxidative stress, and angiogenesis was analysed via qPCR. RO cell density was assessed using immunofluorescence.

Results: Inflammatory cytokine genes, including NFKB1, IL6, and IL1B, were significantly upregulated after exposure to HG and HG with inflammatory factors. SOD1 expression increased after 2 and 4 weeks of 30 mM and 40 mM HG, while SOD2 expression increased with 50 mM HG for 2 weeks. HIF1A expression was significantly higher in 50 mM HG for 2 weeks and 30 mM HG for 4 weeks. These increases were amplified when HG was combined with inflammatory factors. Additionally, cone cell density in RO decreased following 4 weeks of 40 mM HG exposure.

Conclusions: iPSC-derived RPE and RO models effectively mimic key aspects of diabetic retinopathy, making them promising preclinical tools for studying DR pathophysiology and potential therapies.

The impact of allogeneic haematopoietic stem cell transplantation on sickle cell retinopathy and maculopathy: A prospective, observational study

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Purpose: Non-myeloablative allogeneic haematopoietic stem cell transplantation (HSCT) from a matched sibling or haploidentical related donor has become a viable curative therapy for adults with sickle cell disease (SCD). However, data on the effect of HSCT on SCD-related organ complications remain limited. Notably, no studies have investigated the impact of HSCT on ophthalmic complications in adults. We hypothesized that HSCT would halt the progression of pre-existing sickle cell retinopathy (SCR) and maculopathy (SCM) and prevent the development of new ophthalmic complications. To this end, we assessed the incidence of SCR and SCM in adult patients before and at least one year after non-myeloablative matched sibling or haploidentical HSCT.
Methods: Ophthalmic examination, including fundoscopy and macular spectral-domain optical coherence tomography (SD-OCT) and optical coherence tomography angiography (OCTA) scans, was performed in 32 consecutive adults with SCD-(75% HbSS, 9.4% HbSβ^0-thal, 6.3% HbSβ^+-thal and 9.4% HbSC) before and after a median of 28.5 months (range 12–74) post-transplantation. A control cohort of 57 SCD patients not undergoing HSCT was included for comparison.
Results: The progression rates of SCR and SCM post-HSCT were significantly lower compared to the control group comprising all genotypes ($p=0.026$ and $p=0.046$, respectively) and compared to HbSS genotype control patients only ($p=0.049$ and $p=0.047$, respectively).
Conclusions: We conclude that non-myeloablative HSCT can limit the progression of existing or development of new SCR and SCM, a novel finding that expands our understanding of the effects of HSCT on SCD-related organ complications which is crucial for counselling future patients with SCD considering curative therapy.

Characteristics of polypoidal choroidal vasculopathy in Caucasians: A retrospective multicenter cohort study

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Purpose: Polypoidal choroidal vasculopathy (PCV) is often considered a subtype of drusenoid age-related macular degeneration (AMD), but can also exist in patients with central serous chorioretinopathy (CSC). We describe the phenotypes of Caucasian patients with PCV.
Methods: This is a multicenter retrospective cohort study in Amsterdam University Medical Center, Leiden University Medical Center, and The Rotterdam Eye Hospital. The medical charts and multimodal imaging (MMI) of Caucasian PCV patients were assessed retrospectively by two independent assessors, and discrepancies between graders were resolved by a senior retinal specialist. A variety of phenotypical characteristics were graded on MMI: optical coherence tomography, colour fundus photography, fundus fluorescein angiography, and indocyanine green angiography. Based on this grading, PCV patients were distributed among 4 groups: type A: PCV with drusenoid AMD; type B: PCV without drusen but with a BVN; type C: PCV without drusen or a BVN; type D: PCV with CSC.
Results: We included 332 eyes of 305 patients. 126 out of 305 patients were male (41.3%). The average age at diagnosis was 73 years. The following subtypes were found: subtype A in 188 eyes (56.6%); subtype B in 61 eyes (18.4%); subtype C in 15 eyes (4.5%); subtype D in 58 eyes (17.5%). The median BCVA of affected eyes was 0.50 decimal Snellen, with a large spread in each of the groups.
Conclusions: PCV in Caucasians comprises a spectrum of different phenotypes of (chorio)retinal disease. Therefore, we postulate that PCV may occur as a result of different pathophysiological backgrounds.

Distinctive choroidal vascular hyperpermeability patterns in central serous chorioretinopathy correlate with optical coherence tomography angiography parameters: CERTAIN study report #6

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Conflict of interest: No conflicting relationship exists for any author.
Purpose: In central serous chorioretinopathy (CSC), 3 distinct leakage patterns of choroidal vascular hyperpermeability (CVH) on indocyanine green angiography (ICGA) have been previously correlated with disease chronicity by our group: unifocal indistinct signs of choroidal hyperpermeability (uni-FISH); multifocal indistinct signs of choroidal hyperpermeability (multi-FISH); and diffuse indistinct signs of choroidal hyperpermeability (DISH). This report evaluates retinal optical coherence tomography angiography (OCTA) for these CVH patterns in CSC patients.
Patients and Methods: CERTAIN is a monocentric retrospective cohort study of consecutive CSC patients referred to a large tertiary referral center between 01/09/2021 and 30/11/2022. Two independent graders assessed CVH patterns on ultra-widefield (UWF) and 55° ICGA. Eyes were additionally assessed for various OCTA parameters based on 3 mm x 3 mm images taken using the ZEISS PLEX Elite 9000 (Carl Zeiss AC, Jena, Germany).
Results: From 89 CSC patients, 172 eyes were included. From no CVH to uni-FISH to multi-FISH to DISH, vessel density decreased in the retinal deep layer (top of inner plexiform layer to bottom of outer plexiform layer) across the imaged area according to gradings on both UWF and 55° ICGA ($p=0.030$ and $p=0.041$, respectively). Differences were also detected in the inner temporal, inner ring, inner inferior, and average foveal areas.
Conclusions: Vessel density decreases in the deep retinal layer as disease severity increases from no CVH to uni-FISH and multi-FISH to DISH. This suggests that CVH patterns on ICGA may correlate with progressive changes in retinal microvascular perfusion and structural integrity on OCTA.

Distinctive choroidal vascular hyperpermeability patterns in central serous chorioretinopathy correlate with retinal sensitivity levels on microperimetry

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Purpose: In central serous chorioretinopathy (CSC), 3 distinct leakage patterns of choroidal vascular hyperpermeability (CVH) on indocyanine green angiography (ICGA) have been previously correlated with disease chronicity: unifocal indistinct signs of choroidal hyperpermeability (uni-FISH); multifocal indistinct signs of choroidal hyperpermeability (multi-FISH); and diffuse indistinct signs of choroidal hyperpermeability (DISH). This report evaluates the association of these 3 patterns with retinal sensitivity levels on microperimetry in CSC patients.
Methods: CERTAIN is a monocentric retrospective cohort study of consecutive CSC patients referred to a large tertiary referral center between 01/09/2021 and 30/11/2022. Two independent graders separately assessed CVH patterns based on ultra-widefield (UWF) and 55° ICGA. Eyes were assessed for retinal (macular and foveal) sensitivity on microperimetry. A Bonferroni correction was applied to P-values.
Results: From 91 CSC patients, 154 eyes were included. Patients with a DISH pattern showed a lower macular sensitivity compared to uni-FISH reference (−2.61 dB, $p=0.047$) and compared to a combined reference of uni- and multi-FISH (FISH group). Based on 55° ICGA, DISH demonstrated a lower macular sensitivity compared to the uni-FISH reference (−3.13 dB, $p=0.01$) as well as the FISH group (−2.82 dB, $p=0.010$).
Conclusions: The uni-FISH, multi-FISH, and DISH CVH patterns demonstrate association with retinal sensitivity levels in CSC patients, particularly macular sensitivity, which is lower in DISH compared to (uni-)FISH. This highlights the potential role of CVH patterns in assessing disease severity and prognostication in CSC patients.

Comparison of fluorescence molecular imaging using bevacizumab-800CW with fluorescein angiography and acitivity on OCT in neovascular Age-related Macular Degeneration (nAMD)

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Purpose: To determine the correlation between the location of bevacizumab-800CW fluorescence and the area of neovascularization on standard fluorescein angiography (FA). To compare the amount of bevacizumab-800CW fluorescence with the level of activity on OCT in nAMD patients.
Methods: In eight nAMD patients with a type 1 lesion 15 mg of intravenous bevacizumab-800CW was administered. Fluorescence images were taken before injection and after 3–4 days. Contrast-to-noise ratios (CNRs) were calculated for areas that showed visible fluorescence accumulation during the follow-up visit.
FA-ROI CNR of the bevacizumab-800CW images was calculated in the area of neovascularization on FA. The activity of the neovascularization on OCT was categorized as mild or severe based on the amount of sub- or intraretinal fluid.
Results: Five patients on anti-VEGF treatment and three treatment naïve patients were included. One of these patients was excluded due to poor image quality.
The CNR of bevacizumab 800CW correlated well to the FA-ROI CNR of standard FA; correlation coefficient 0.94.
The CNR of bevacizumab-800CW images was higher (average CNR 6.63) in patients categorized as having severe activity on OCT compared to patients categorized as having mild activity on OCT (average CNR 3.39).
Conclusions: The fluorescence of bevacizumab-800CW correlated well to the location of the neovascularization on standard fluorescein angiography and the amount of bevacizumab-800CW fluorescence corresponded to the activity of the neovascularization on OCT.

Multimodal imaging characteristics after subretinal fluid resolution in central serous chorioretinopathy

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Purpose: To compare choroidal and retinal characteristics in patients with central serous chorioretinopathy after spontaneous resolution of subretinal fluid (SRF) and after SRF resolution following half-dose photodynamic therapy (HD-PDT).
Methods: In this retrospective study, 46 patients with spontaneous resolution were compared with 46 age- and gender matched patients with resolution after HD-PDT. Optical coherence tomography (OCT), indocyanine green angiography (ICGA), and fluorescein angiography (FA) were performed at the first visit after SRF resolution. For the HD-PDT group, imaging was performed at a median of 7.5 weeks after HD-PDT. On OCT, we analysed the central foveal thickness (CFT), subfoveal choroidal thickness (SFCT), ellipsoid zone (EZ) continuity, and the presence of pigment epithelial detachments (PEDs). On indocyanine green angiography (ICGA) and fluorescein angiography (FA), we evaluated the presence and the area of hyperfluorescence.
Results: Regarding the choroid, the mean SFCT was thicker ($418.9 \pm 84.9\,\mu m$ vs. $353.4 \pm 88.6\,\mu m$, $p = 0.006$) and the median hyperfluorescent area on ICGA was larger ($1.10 mm^2$ [0.32–2.39] vs. $0.44 mm^2$ [0.11–0.99], $p = 0.002$) in the spontaneous resolution group when compared with the HD-PDT group. At the level of the neuroretina and retinal pigment epithelium, we observed no difference in the hyperfluorescent area on FA (i.e., window defects), CFT, and presence of PEDs between the two groups. The EZ continuity was higher after HD-PDT ($p = 0.002$).
Conclusions: Patients with spontaneous resolution had more ICGA hyperfluorescence and a thicker choroid than patients treated with HD-PDT. The favourable choroidal changes after HD-PDT might explain why recurrences occur less frequently after HD-PDT.

Evaluating systemic complement profiles in patients with age-related macular degeneration

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Purpose: Characterizing systemic complement profiles in patients with age-related macular degeneration (AMD).
Methods: AMD patients with available complement measurements were included. Complement factors (H, I, B, D, C2, C5, properdin) and activation markers (Bb, C3bBbP, C3a, C3bc, C5a, TCC, C4d, C4d/C4 ratio) were measured in plasma. Screening for rare or low-frequency variants in complement genes was performed. Patients were classified based on phenotype (**severe or non-severe**) and complement profile (**extreme or non-extreme**). Severe phenotypes were defined as intermediate or advanced AMD in patients >35 years or any AMD in patients ≤35 years. Extreme complement profiles were defined as FH, FI, and/or activation marker levels >2 standard

deviations above or below the mean or as clear outliers in plots.
Results: Sixty-six patients (mean age 58.4 years) were included. Ten patients (15.2%) had extreme complement profiles. Of these, seven (70%) had a severe phenotype and three (30%) had a non-severe phenotype. The most extreme values were found in FH (30%) and in both C3bBbP and TCC (30%). Of the remaining 56 patients with non-extreme complement profiles, 44 (66.7%) had severe phenotypes, and 12 (18.2%) had non-severe phenotypes. One patient with a severe phenotype stood out due to very high activation markers, despite not carrying rare or low-frequency variants in complement genes.
Conclusion: Systemic complement profiles in AMD patients are variable, with extreme complement corresponding with extreme phenotypes in only 70% of patients. This highlights the need for large-scale studies, investigating complement profiles and their potential role in guiding therapeutic strategies for AMD.

Genetic characterization of patients with cuticular drusen and pseudovitelliform lesions

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Purpose: To genetically characterize patients with cuticular drusen (CD) and pseudovitelliform lesions (PVL).
Methods: Patients with CD and PVL were retrospectively collected from our age-related macular degeneration (AMD) database, confirming diagnosis with multimodal imaging reviewed by two retinal specialists. We screened the coding regions and splice-sites of the *CFH*, *CFI*, *CFB, C3, C2, C5*, *CFHR1-5*, *CFP*, and *VTN* genes for rare and low-frequency variants, classified based on ACMG-AMP criteria. Genetic risk scores (GRS) were calculated based on 52 AMD-associated risk variants and categorized as low, intermediate, or high. Risk variants in *ARMS2* (rs3750846) and *CFH* (rs1061170) were also assessed independently.
Results: Sixteen patients (mean age 58.9) were included. All had bilateral CD, 11 had bilateral PVL, and 5 had unilateral PVL. Mean GRS was 1.02±1.38 (intermediate), which is comparable to the GRS of patients with intermediate AMD. Minor allele frequencies of the risk variants in *CFH* and *ARMS2* were 0.406 and 0.156, respectively, which is lower than expected in an AMD population. In contrast, four rare or low-frequency variants (3 in *CFI* and 1 in *CFHR4*) were identified in three patients (18.8%), which is much higher compared to a normal AMD population. Misdiagnosis with an inherited retinal dystrophy was ruled out by whole exome sequencing.
Conclusion: We found that genetic risk factors in patients with CD and PVL overlap with those associated with AMD, but there are some differences. Strikingly, the frequency of rare variants, especially in the *CFI* gene, is much higher in this subgroup of patients.

Geographic atrophy in patients with age-related macular degeneration is associated with rare variants in complement factor H and complement factor I

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Purpose: To describe the phenotype of patients with age-related macular degeneration (AMD) carrying rare genetic variants in the complement factor H (*CFH*) and complement factor I (*CFI*) genes.
Methods: 234 AMD patients carrying rare coding and splice-site variants in *CFH* and *CFI*, and 234 AMD noncarriers from the European Genetic Database were included. Phenotypic characteristics on colour fundus photographs were graded according to the Rotterdam Classification and compared between carriers and non-carriers by univariate generalized estimating equations binary logistic regression analyses with a Bonferroni correction for multiple comparisons. We performed sub-analyses for pathogenic rare variants only, and we analysed *CFH* and *CFI* carriers separately.
Results: Disease stage (geographic atrophy (GA) and intermediate AMD), predominant drusen type, largest drusen size and drusen area were associated with carriership of rare pathogenic variants in *CFH* ($p<0.001$, $p=0.002$, $p<0.001$ and $p<0.001$, respectively). Disease stage (GA and intermediate AMD), drusen size, drusen area and retinal pigment epithelium pigmentation were associated with carriership of rare pathogenic variants in *CFI* ($p=0.01$, $p=0.006$, $p<0.001$ and $p=0.006$, respectively). Carriers of rare pathogenic variants in *CFH* were younger ($p<0.001$) and had a lower genetic risk score for common AMD-associated variants compared to noncarriers (mean (SD) GRS 0.83 (1.01) vs. 1.41 (1.21), $p=0.03$).
Conclusions: AMD patients carrying rare variants in *CFH* and *CFI* had a more severe drusen phenotype and a higher frequency of GA at a relatively early age. Identifying this distinct phenotype could aid in pinpointing individuals who are more likely to benefit from complement inhibiting therapies.

One-year results of a randomized controlled trial for lifestyle intervention in age-related macular degeneration: AMD-Life

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Purpose: Lifestyle changes can reduce the risk of progression in Age-related Macular Degeneration (AMD) by 50%, but adherence to recommendations is often low. AMD-Life is a randomized controlled trial addressing this challenge. We present the first-year results of the primary outcome: the Total Lifestyle Score (TLS).
Methods: 150 AMD patients were randomized into three groups: (A) standard recommendations; (B) standard recommendations plus personalized risk profiling based on genetics and lifestyle; and (C) group B plus additional coaching using behavioural change techniques (BCT) and motivational interviewing (MI). Lifestyle and diet data were assessed at months 0, 3, 6, 9, and 12 through comprehensive questionnaires. TLS, a 13-point score, measured adherence based on non-smoking, BMI <25, Mediterranean diet, and physical activity. Changes in TLS were analysed using Wilcoxon signed-rank tests.
Results: Among 150 participants (A: $N=49$, B: $N=52$, C: N=49), 3 (2.0%) were lost to follow-up, and 11 (7.3%) dropped out (4 in A, 6 in B, 5 in C). After 1 year, TLS in group A remained unchanged (7 [5–8] to 7 [6–8], $p=0.35$); group B improved from 6 [5–7] to 7 [5–8] ($p=0.027$), and group C showed significant improvement from 6 [4–7] to 8 [6–10] ($p<0.001$).
Conclusions: Lifestyle changes are feasible in AMD patients, but significant improvements occur only with personalized risk assessments and coaching. These findings could shift clinical strategies to effectively promote risk-reducing behaviour in AMD.

Radiation retinopathy: Insights into development and treatment

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Introduction: Since 1999, uveal melanoma at Erasmus MC have been treated with eye-sparing radiotherapy, including fractionated stereotactic radiotherapy (fSRT) using XKnife and Cyberknife (since 2016), and proton beam radiotherapy (PBR, introduced in 2020). A common complication is radiation maculopathy (RM) and retinopathy (RP), often resulting in significant vision loss. However, treatment strategies for these complications remain underexplored with limited long-term studies.
Patients and Methods: This retrospective study analysed the onset of RP/RM and evaluated treatment approaches. PBR-treated patients received preventive intravitreal Avastin shortly after radiation, whereas fSRT patients were treated only when indicated. We assessed 210 XKnife-treated, 23 Cyberknife-treated, and 58 PBR-treated patients, focusing on the median time to RP/RM onset, tumour size, and treatment interventions.
Results: RP/RM occurred in 41.9% of XKnife patients (median onset: 24.7 months; median tumour diameter: 11.5 mm; thickness: 5.5 mm), 32.5% of Cyberknife patients (19.8 months; 11.7 mm; 5.7 mm), and 19% of PBR patients (median onset: 8.7 months; 13.1 mm; 6.9 mm). Avastin was administered to 73% of PBR patients (started preventive), compared to 43% and 24% in the Cyberknife and XKnife groups (first treatment was in 2008), respectively, at time of RP/RM. PRP was performed in 17% in XKnife versus 5% in Cyberknife groups, and 9% of PBR patients.
Conclusion: Preventive Avastin following PBR showed a lower incidence of RP/RM. However, improved imaging and evolving treatment strategies have also facilitated earlier detection and intervention in recent years.

Evaluating the 'Macular Degeneration: What Happens Next?' Question Prompt List: Enhancing communication in ophthalmology consultations

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Purpose: Effective communication is essential in healthcare, as it fosters trust between patients and providers, enhances comprehension of diagnoses and treatment plans, and mitigates patient anxiety and stress levels. Age-related Macular Degeneration (AMD) is a complex condition where effective communication between patients and healthcare providers (HCPs) plays a vital role. This study evaluates the communication tool 'Macular Degeneration: What Happens Next?', a Question Prompt List (QPL) designed to help patients formulate questions and assist HCPs in guiding conversations.
Methods: The development process included a literature review and surveys to gather insights into patients' experiences, needs, and preferences. These findings guided focus group discussions with patients, informal caregivers, and HCPs to shape the communication tool. The

QPL was further refined through cognitive interviews. The tool's impact is being evaluated using video recordings of consultations between patients and ophthalmologists (n=51). Post-consultation surveys assess patients' experiences with communication, information provision, and shared decision-making, and evaluate the QPL.
Results: Initial feedback highlights the QPL's acceptability and potential impact. Early results suggest that 3 out of 4 patients are willing to use the tool, and all recommend it to others with AMD. The tool received positive feedback with an average rating of 7.6 out of 10. Analyses of video consultations and survey data provide further insights into its role in enhancing patient-provider communication.
Conclusions: The QPL demonstrates potential to enhance patient-provider communication. It may further provide a foundation for creating similar tools for other ophthalmic conditions.

Diminished plasma serotonin and taurine levels are a metabolic feature of age-related macular degeneration

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Purpose: This study aimed to compare metabolic profiles of peripheral blood plasma in early and late age-related macular degeneration (AMD) compared to sex- and age-matched controls.
Methods: Peripheral blood plasma samples were collected from participants with early AMD (n=21), late AMD (n=39), and controls (n=39). Metabolomic profiling was performed using a 619-plex targeted array with liquid chromatography– and flow injection analysis tandem mass spectrometry. AMD classification followed Age-Related Eye Disease Study (AREDS) criteria. The AMD Genetic Risk Score (GRS-52), previously calculated using 52 disease-associated SNPs, was also used for statistical analyses in this study. Metabolite levels were analysed using regression models adjusted for age, sex, and smoking status.
Results: No significant associations were found between metabolites and the GRS-52. However, we identified significant changes in 3 small molecules, 5 lipids, and 5 metabolic indicators in late AMD versus controls. Plasma serotonin and taurine levels showed the strongest reductions ($p=1.03\times10^{-13}$; $p=2.36\times10^{-9}$, respectively), with their concentrations progressively declining as the disease advanced. Analysis of predefined sums and ratios of metabolites (metaboINDICATOR™) confirmed decreased serotonin and taurine synthesis in late AMD ($p=4.41\times10^{-14}$; $p=2.3\times10^{-29}$). These changes were unrelated to serotonin reuptake inhibitor use.
Conclusions: AMD is characterized by progressive metabolic alterations, including significant depletion of serotonin and taurine in peripheral blood plasma. These findings highlight potential biomarkers and therapeutic targets for AMD progression.

Two-year follow-up of dutch patients with *RPE65*-associated inherited retinal dystrophy: Clinical outcomes following treatment with voretigene neparvovec

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Purpose: To report clinical outcomes of gene therapy with voretigene neparvovec (VN) in a large cohort of patients with inherited retinal dystrophy caused by biallelic *RPE65* mutations.
Methods: Analysis of clinical records, ancillary testing, and multimodal retinal imaging of patients treated with voretigene neparvovec (VN) in a multi-center, prospective, observational study. Efficacy was evaluated using outcome measures including scotopic full-field stimulus threshold testing (FST), best-corrected visual acuity (BCVA), low-luminance visual acuity (LLVA), semi-automated kinetic perimetry (SKP), and microperimetry.
Results: Twenty-three patients (n=36 eyes) treated between 2021 and 2023 were included for analysis, of whom 19 patients (n=31 eyes) completed the 24-month follow-up. Postoperative retinal sensitivity, measured with FST, was improved by 1 month of treatment (2 log cd.s/m^2). BCVA did not show improvement, while LLVA demonstrated significant gains (0.41 LogMAR). SKP was highly variable without a clear trend. Mean retinal sensitivity, assessed using microperimetry, was consistently measurable in 6 eyes and showed improvement at 24 months (3.24 dB). Progressive chorioretinal atrophy observed

in almost all eyes has not adversely affected visual outcomes following VN treatment for at least 24 months.
Conclusion: Data on VN in a clinical setting show sufficient effectiveness of VN with persistence of improvements in retinal sensitivity and LLVA up to 24 months. The long-term effect of progressive chorioretinal atrophy on visual function will require ongoing monitoring.

Ultra-high-speed, ultra widefield optical coherence tomography angiography of peripheral retinal vascular abnormalities

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Purpose: To compare in vivo examination of peripheral retinal vascular abnormalities on fluorescein angiography (FA) with ultra high-speed, ultra widefield optical coherence tomography angiography (OCTA).
Methods: The eyes of patients with diabetic retinopathy and vascular occlusions were examined using the Heidelberg Engineering Spectralis™, the Optos California™ and the TowardPi BMizar™. FAG recordings were made with the first two devices and compared with OCTA recordings from the latter. The clinical information of the images was compared with each other.
Results: We included 15 images of patients with retinal vasculopathy. In each patient, the vascular situation, including non-perfusion areas and the underlying choroidal vessels, could be properly assessed to the periphery using ultra high-speed, ultra widefield OCTA. FA performed comparable detecting ischemia up to midperiphery, however, the extreme periphery could only be assessed with the Optos™ system and the images were often less clear than with OCTA. Leakage and staining could only be assessed by FA and retinal thickening only by OCTA.
Conclusion: In this study, ultra-high-speed, ultra widefield OCTA appeared to be a suitable technique to noninvasively record retinal vascular abnormalities and edema to the periphery. Leakage and staining could only be detected using FA, but very peripheral areas of the retina were often more difficult to assess than with OCTA.

Treatment intensity in retinal vein occlusion: 24-month analysis of FRB! practitioners

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Purpose: To demonstrate 24-months outcomes of eyes managed by practitioners benchmarked using a funnel plot by their frequency of treatment using vascular endothelial growth factor (VEGF) inhibitors for naive retinal vein occlusion (RVO).
Methods: Multicentre, international, observational study of 29 practitioners in 12 countries managing 1110 eyes with RVO commencing VEGF inhibitors between January 1, 2012–2022 tracked in the Fight Retinal Blindness! registry.
Results: We identified 3 outlying "intensive" practitioners (managing 350/1110 eyes [32%]), 22 "typical" practitioners (604/1110, [54%]) and 4 outlying "relaxed" practitioners (156/1110, [14%]). Their respective 24-month outcomes in Branch and Central RVO included firstly mean adjusted change in Visual Acuity (VA) in BRVO: +16.2, +13.6, +9.3 letters ($p<0.01$) and CRVO: +14.2, +12.7, +4.8 letters ($p<0.01$), secondly adjusted change in macular thickness in BRVO -179, −150, −159 µm ($p<0.01$) and CRVO -324, −283, −232 µm ($p<0.01$), furthermore median injections 18, 13 and 10, and lastly median final injection intervals, BRVO 6, 9, 10 weeks and CRVO 6, 9, 12 weeks. There were no significant differences in adverse outcomes between the practitioners.
Conclusion: At 2 years, the intensive practitioners treated RVO patients using VEGF inhibitors with nearly twice the frequency of the relaxed practitioners however their patients had gained twice (BRVO) to three times (CRVO) more letters of visual acuity. The outcomes of the intensive practitioners suggest that RVO patients receive under-treatment in routine care.

Using continuous visual stimulus tracking for detecting visual function loss due to acquired brain injury

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Introduction: Acquired brain injury (ABI) can lead to visual field (VF) defects. Standard automated perimetry (SAP) is the gold standard for VF assessment, but performing SAP can be too demanding for some individuals. We previously demonstrated that a much simpler task involving tracking a continuously moving stimulus enables detecting and quantifying glaucomatous visual function loss. This study investigated whether we can do so similarly for ABI.
Patients and methods: Tracking performance (agreement between gaze and stimulus positions) was evaluated in 16 cases with homonymous VF defects from ABI (median [IQR] age 66 [62–68] years); 8 with right and 8 with left hemisphere lesions. This data was compared to previously collected data of 36 controls (70 [67–72] years) and

36 glaucoma cases (70 [67–74] years). All participants monocularly tracked a moving stimulus (Goldmann size-III) at three Weber contrast levels (40–160–640%). Their eye movements were recorded during the task.
Results: The presence of ABI significantly reduced tracking performance compared to controls ($p<0.001$), independent of lesion side ($p=0.53$). Performance improved from 40% to 160% contrast ($p<0.001$), but plateaued at 640% contrast ($p=0.53$). Tracking performance did not well explain the extent of visual function loss (in terms of SAP Mean Sensitivity) ($R^2<0.36$). Strikingly, ABI cases with comparable function loss to glaucoma cases showed less reduction in tracking performance, and a smaller increase in tracking performance with increasing contrast.
Conclusions: Visual function loss from ABI results in decreased tracking performance, but the reduction is much smaller than in glaucoma.

Characteristics of autosomal dominant *WFS1*-associated optic neuropathy and its comparability to *OPA1*-associated autosomal dominant optic atrophy

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Introduction: *WFS1*-associated diseases, or 'wolframinopathies', exhibit a spectrum of ocular and systemic phenotypes, of which the autosomal recessive Wolfram syndrome has been the most extensively studied. The most common phenotype in autosomal dominant (AD) *WFS1*-associated disease, the combination of OA and hearing loss (HL), clinically resembles the 'plus' phenotype of dominant optic atrophy (DOA), caused by mutations in the *OPA1*-gene. This study aims to describe the ophthalmic characteristics of AD *WFS1*-associated optic atrophy (AD *WFS1*-OA), and to explore phenotypic differences with DOA.
Materials & Methods: A medical record review was performed for genetically confirmed AD *WFS1*-OA cases in six tertiary ophthalmic centers in The Netherlands and one in Belgium. Demographic, ophthalmological and genetic data were collected. The findings were compared to age matched DOA patients with confirmed *OPA1* mutations.
Results: Eighteen patients (82%) had HL in addition to OA. Diabetes mellitus was found in 7 (32%). Four patients had isolated OA. One patient had an unusual phenotype with anterior chamber abnormalities and malformations of the extremities. Compared to DOA, AD *WFS1*-OA patients had different colour vision abnormalities (red-green vs. blue-yellow in DOA), abnormal OPL lamination on macular OCT (absent in DOA), more generalized thinning of the retinal nerve fibre layer, and more reduced and delayed pattern reversal visual evoked potentials.
Conclusions: AD *WFS1* mutations have a broad phenotypic spectrum, that may include isolated optic atrophy. Colour vision testing, macular and peripapillary OCT, and assessment of pattern reversal VEP can aid in clinically distinguishing *WFS1*-OA from DOA.

Measurement of the optic nerve sheath diameter (ONSD) by ultrasonography in patients with increased intracranial pressure

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Background and purpose: Assessment of patients with suspected raised intracranial pressure (ICP) include a variety of modalities. Measurement of the optic nerve sheath diameter (ONSD) with ultrasonography is a non-invasive diagnostic modality. This study investigates reliability of using ONSD measurement as a non-invasive modality in assessment of suspected and raised ICP.
Methods: A retrospective study. Following an ocular examination, a transorbital ultrasonographic (USG) cross section of the retrobulbar optic nerve (ONSD measurement) was obtained in symptomatic 144 patients using a 10-MHz linear array transducer.
Results: The mean age was 25.42 years. 61of the patients were males (42.06%) and 83 of the patients were females (57.42%). The most common diagnosis was idiopathic intracranial hypertension (40 [27.77%]).
In 48 patients (33.33%) lumbar puncture was performed. 38 of the 48 patients (79.16%) had a high opening pressure. 13 of the 48 patients (27.08%) had a high ONSD in at least one eye. 20 of the 48 patients (41.66%) had a swollen disc in at least one eye. 27 out of 48 (56.2%) patients had either an abnormal ONSD or swollen disc.

Conclusion: Measurement of ONSD using USG could possibly aid in the assessment of patients with raised ICP. However, ONSD should not be used solely in the diagnosis, and it is very important to realize that patient can have raised intracranial pressure despite have a normal appearance of the optic disc and a normal ONSD measurement.

Immunosuppression or peptide receptor radionuclide therapy in xanthogranulomatous disease?

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Background: Adult orbital xanthogranulomatous disease is rare. This group of disorders represents a spectrum of non-Langerhans cell histiocytosis with characteristic histopathology. Four subtypes are known: adult onset xanthogranuloma (AOX), adult onset asthma and periocular xanthogranuloma (AAPOX), necrobiotic xanthogranuloma (NBX) and Erdheim Chester disease (ECD).
Methods: Review of our series of patients diagnosed with this condition and treated in Rotterdam.
Results: Debulking proved often sufficient for patients with AOX Patients with AAPOX, NBX and ECD frequently required systemic therapy, including steroids + azathioprine In ECD treatment with peptide receptor radionuclide therapy resulted in remission of disease for over 17 years.
Conclusion: Peptide receptor radionuclide therapy is a novel treatment for ECD, but further study in all subtypes is needed.

Characteristics of patients with craniofacial fibrous dysplasia and optic neuropathy

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Purpose: To characterize patients with craniofacial fibrous dysplasia and optic neuropathy.
Methods: From our cohort of 364 patients with fibrous dysplasia / McCune Albright syndrome (FD/MAS), we selected the patient files of 179 patients with craniofacial fibrous dysplasia (CFD). We identified 13 patients with optic neuropathy (ON) and extracted the following clinical characteristics: cause of ON, age of onset, treatment, response to treatment and available data on visual acuity, visual fields, ocular coherence tomography (OCT) data of the papillary retinal nerve fibre layer (pRNFL) and macular ganglionic cell layer (mGCL).
Results: Median age of onset of optic neuropathy was 17 years (5–31 years). Compression of the optic nerve was present in 11 eyes because of narrowing of the optic canal, four optic nerves were compressed by an aneurysmal bone cyst (ABC) and two optic nerves had iatrogenic optic nerve damage after surgery not related to the optic nerve. All patients with ABC were decompressed surgically and visual acuity and visual fields recovered fully. Patients with narrowed optic canal and no treatment were either stable or worsened, whereas the decompression surgery led to visual recovery in four cases, stable visual function in two and worsening in one patient.
Conclusion: Optic neuropathy in patients with craniofacial fibrous dysplasia is more likely to recover after surgery when it is caused by ABC then by narrowing of the optic canal. Patients should be monitored on regular intervals and should receive instructions on rapid onset visual deterioration.

Treating socket discharge and discomfort: A crossover randomized trial of vitamin a ointment, low-dose steroids eyedrops and artificial tears

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Purpose: This study evaluates the effects of artificial tears, vitamin A ointment, and low-dose steroid drops on chronic irritation and discharge in anophthalmic patients wearing ocular prostheses. It also aims to identify patient subgroups that may benefit from specific regimens.
Methods: This prospective, crossover randomized trial followed a Williams design with 40 participants aged ≥16 years who wore ocular prostheses and experienced chronic discharge or discomfort. Each participant underwent four 2-week intervention periods (artificial tears, vitamin A ointment, low-dose steroids, and no medication as control), separated by 2-week washout periods. Visual analogue scale questionnaires assessed discharge and irritation frequency and severity. Data were analysed using linear mixed models with Bonferroni correction for multiple comparisons, and subgroup analyses were descriptive.
Results: All three regimen were preferred equally by the patients (1/3vitamin A ointment, 1/3articial tears, 1/3 FML), but vitamin A ointment most consistently improved all outcome parameters: irritation frequency (−1.766, 95%CI [−2.857, −0.675], $p<0.001$), irritation severity (−1.448, 95%CI [−2.564, −0.332], $p=0.004$), discharge frequency (−1.186, 95%CI [−2.295, −0.077], $p=0.029$), and discharge severity (−1.146, 95%CI -[−2.177, −0.114], $p=0.021$). Artificial tears significantly reduced irritation frequency (−1.279, 95%CI [−2.370, −0.188], $p=0.013$), FML showed both very positive, and sometimes negative effects.. Patient subgroups with lagophthalmos or dry

anophthalmic socket syndrome did not show to benefit from a specific regimen.

Conclusions: Vitamin A ointment is the most effective intervention, followed by artificial tears, for managing chronic irritation and discharge in anophthalmic patients wearing ocular prostheses.

Diffusion restriction in IgG4 related orbital disease

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Purpose: In Orbital inflammation MRI Diffusion Weighed Imaging (DWI) does not show diffusion Restriction (DR), while in orbital lymphoma it does. The purpose of this study was to determine DR in IgG4 Related Orbital disease (IgG4-ROD) which is an inflammatory disease.

Methods: In this study we retrospectively studied 13 patients with histologically proven IgG4-ROD (23 orbital lesions) We extracted the following clinical characteristics: age, gender, location of the orbital inflammation, IgG4 level in serum presence and presence of other localizations of IgG4 related disease (IgG4-RD) and we determined DWI and ADC of the orbital process.

Results: Mean age was 61 years, 6 males and 7 females, 17 locations of IgG4 ROD were in the lacrimal gland, 6 elsewhere in the orbit. Four patients had localization of IgG4-RD elsewhere in the body. Mean serum IgG4 was 1.95 g/L, in 8 patients it was elevated (>0.864 g/L). Sixteen locations of IgG4-ROD showed DR (70%), with mean ADC 0.88, which is indicative for a malignancy.

Conclusions: IgG4-ROD shows DR with ADC values indicating a likely malignant process. More study is needed to explain this and to investigate the consequences.

Bilateral spheno-orbital meningioma: A rare entity? Ten new cases

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Introduction: To define bilateral spheno-orbital meningioma (SOM) and describe clinical and radiological features.

Methods: We defined bilateral SOM as: bilateral (1) dural involvement located at the sphenoid ridge, and (2) hyperostosis of the sphenoid bone and (3) orbital involvement, all three per side continuous. The electronic files of all SOM patients seen at our multidisciplinary outpatient clinic were reviewed. MRIs and CTs were evaluated by a neuroradiologist, an orbital surgeon and a neurosurgeon.

Results: Images of 127 patients were evaluated and ten patients with bilateral SOM were identified, all female, aged at diagnosis between 41 and 62 years old (average 49.6). One patient was diagnosed with neurofibromatosis type 2 and two with meningiomatosis. There was a history of severe depression in three patients and three others were intellectually disabled. All patients had a history of contraceptive medication and at least seven patients had a history of injectable medroxyprogesterone usage. There was compressive optic neuropathy in 10 out of 20 eyes, bilateral in four patients. Four patients underwent bilateral surgery and three patients unilateral surgery. Two patients needed radiotherapy. Average follow-up was 6.6 (3–13) years. Visual acuity improved from 0.7 to 1.4, $p=0.09$, visual field index also improved from 70% to 95%, $p=0.028$, proptosis remained stable (average 1 mm difference).

Conclusion: Bilateral SOM might not be as rare as reported in literature. There is a possible relationship with use of progestins. Treatment strategy and indications for simultaneous bilateral surgery depend on the presence of optic neuropathy and clinical features.

Muscle elongation surgery in patients with Graves' Orbitopathy (GO): A prospective multicenter study

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Objective: Patients with severe ocular motility restriction and/or large angle strabismus due to Graves' orbitopathy (GO) cannot be adequately addressed with large muscle recessions. Muscle tendon elongation surgery, using a spacer of bovine pericardium has emerged as a valuable alternative. The purpose of this study is to assess the outcome of muscle elongation surgery in patients with GO.

Methods: Prospective observational cohort study in 3 centers. All GO-patients who needed elongation of one or more rectus muscles were included. One week preoperative and 3 and 6 months postoperative, ophthalmic and orthoptic examination, ductions, field of binocular single vision (BSV) and GO-Quality of Life (QoL) were assessed.

Results: 35 Patients were included in the study (20 females, 15 males, mean age 60 years). Preoperative horizontal deviation was 45 prism diopters (PD) and the vertical deviation 35 PD. The mean length of spacer used was 8.2 mm. Muscle tendon elongation was performed on 15 inferior rectus muscles, 33 medial rectus muscles

and 8 superior rectus muscles. Duction range remained stable. Nine patients required a second surgery, all without muscle elongation.
BSV improved from 0 points preoperatively to 74 points postoperatively. QoL improved from 47 points to 94 points.
Conclusion: Using Tutopatch in GO patients with large angle and/or severely restricted strabismus, ductions remain stable and a large field of BSV can be achieved with associated improvement of the QoL.

The effect of oblique muscle surgery in children with non-syndromal unicoronal craniosynostosis

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Purpose: To determine the effect of oblique muscle surgery on ocular alignment in children with non-syndromal unicoronal craniosynostosis (N-UCS) presenting with a V-pattern and oblique muscle dysfunction.
Methods: We retrospectively reviewed all N-UCS children (N=76) treated for V-pattern strabismus with oblique dysfunction between 1992 and 2024 at our clinic. Children were included if they had undergone oblique muscle surgery only. Changes in ocular alignment in primary position (PP), elevation in adduction, and V-pattern were assessed.
Results: Forty-nine children were treated with oblique muscle surgery. Mean age at surgery was 5.7 yrs. (range 1.5–12.1 yrs). 33 surgical procedures could be analysed: 19 inferior oblique recessions (IOR), 5 IOR with contralateral inferior oblique partial tenotomy (IOR+IOT), 5 IOR with superior oblique tuck (IOR+SOT), 2 IOT, 1 SOT, and 1 inferior oblique disinsertion with SOT. For IOR, median vertical deviation improved from 5.5°PD to 1°PD, and V-pattern from 7°PD to 2°PD. In IOR+SOT, vertical deviation in PP reduced from 8.5°PD to 3.5°PD, and V-pattern from 15°PD to 5°PD. For IOR+IOT, vertical deviation improved from 8.5°PD to 5°PD, and V-pattern from 8°PD to 4°PD. Pre-operatively, torticollis was present in 19 patients and improved after surgery in 79%. Satisfaction with surgical outcome was reported by 72%.
Conclusion: In N-UCS patients oblique muscle surgery was successful in improving a vertical deviation, V-pattern, and torticollis. Findings of this study emphasize that conducting oblique muscle surgery in this group of children should not be assumed to have a limited effect.

Abducens palsy caused by neurovascular conflict at the root-exit zone

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Introduction: an acquired abducens nerve palsy at a later age has many causes, the aetiology of which can usually be assessed by the anamnesis. A slowly progressive abducens nerve palsy can be caused by compression in the course of the cranial nerve. The clinical presentation is with horizontal uncrossed diplopia, torticollis rotation to ipsilateral side, esotropia that is greater at distance than at near, and a limitation of abduction of the affected eye.
Method: we describe three patients with slowly progressive horizontal diplopia and esotropia who were initially helped with prism glasses, which were later insufficient. On radiological evaluation by means of MRI, an abnormality was found in all three at the location of the brainstem, where the abducens nerve exits.
Results: patient 1 (59-year-old man) already had a bimedial recession of the m rectus medialis at the age of 50. His eye position was 40^ esotropia OS at far, with a significant abduction restriction OS not beyond the midline. After Hummelsheim transposition, the patient was orthotropic without diplopia. The abduction restriction remained. Patient 2 (68-year-old man) had horizontal diplopia at far of 16^ and a mild abduction restriction OS. A 6 mm recession m rectus medialis reduced the angle to 6^ esotropia far 1 week postoperatively; long-term follow-up was suspended due to an unrelated problem. Patient 3 (64-year-old male) had 20^ esotropia at a distance with mild abduction restriction OS. A 7 mm recession m rectus medialis OS improved the eye position to 6^ esophoria at a distance without diplopia. In all, there was radiological evidence of a neurovascular conflict with the n. abducens and the a. cerebelli inferior anterior or the a. basilaris. Theoretically, microvascular decompression via the neurosurgeon should be possible, but no literature is known about this yet.
Conclusions: these patients illustrate that compression of a vessel on the n. abducens at the brainstem can be a cause of a slowly progressive abducens paresis.

Results of the modified Nishida procedure in unilateral sixth nerve palsy

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Introduction: To evaluate the outcomes of the modified Nishida procedure in patients with abducens nerve palsy, particularly in those with poor or absent lateral rectus function.
Methods: We conducted a retrospective review of 7 patients (4 male, 3 female; mean age 57.1 ± 22 years)

diagnosed with abducens nerve palsy who underwent the modified Nishida procedure. This technique involves transposing the lateral margin of each vertical rectus muscle superotemporally or inferotemporally and suturing them to the sclera, without muscle tenotomy or splitting. Preoperative, surgical, and postoperative data were analysed, including deviation, abduction, and head turn.
Results: Five patients underwent combined vertical rectus transposition and ipsilateral medial rectus recession. Preoperatively, the mean deviation was 27.6 ± 13.6 PD (range: 16–56) and mean abduction was −8 ± 11 degrees (range: −19 to +12). At 8 weeks postoperatively, the mean deviation improved to 6.2 ± 5.8 PD (range: −4 to +14), and mean abduction was −8 ± 12.8 degrees (range: −20 to +10). Head turn was completely resolved in 5 patients, with 2 showing minimal residual head turn.
Conclusion: The modified Nishida procedure is a safe and effective treatment for abducens nerve palsy, offering significant improvement in ocular alignment and partial restoration of abduction. All patients experienced resolution of diplopia and significant head turn, with no surgical complications. This minimally invasive approach is particularly beneficial for patients at risk of anterior segment ischemia.

Through the eye of the painter, the disadvantages of stereopsis

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Purpose: One of the challenges for a painter is to create depth in a flat surface. Painters who do not have depth perception may have an advantage over painters who do have depth perception.
Methods: Literature review of ophthalmological abnormalities in painters who have produced a painting that has been sold for more than 100 million dollars.
Results: 22 painters were identified. Ophthalmological abnormalities are frequently described in these painters, with an exotropia occurring above average.
Conclusion: Exotropia may be an advantage for a painter to create depth in a flat surface.

Uveitic macular edema in ocular tuberculosis patients in a non-endemic country: Characteristics, management, and visual outcomes

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Objective: To describe clinical features, treatment strategies, and visual acuity changes of eyes with uveitic macular edema (UME) in ocular tuberculosis (OTB) patients from a non-TB-endemic country.
Methods: This retrospective study was conducted using a 10-year period registry of OTB patients diagnosed in Erasmus MC, Rotterdam. Longitudinal analysis of visual acuity trajectory in eyes with and without UME was performed using linear mixed effect model.
Results: Out of 93 included patients, 23 (24.7%; 26 eye episodes) presented with baseline UME. Older age ($p = 0.024$) and diabetes coexistence ($p = 0.048$) were associated with UME. Eyes with baseline UME showed lower presenting best-corrected visual acuity (BCVA) ($p = 0.024$). Posterior uveitis ($p = 0.005$), the presence of active vitreous cells ($p = 0.016$), and retinal vasculitis ($p = 0.008$) were ocular signs associated with UME. A step-wise treatment approach primarily initiated with local steroids, followed by a combination with oral acetazolamide and, if necessary, additional systemic immunosuppressants. Overall, this approach resulted in complete UME resolution in 77% (20/26) of cases. UME resolution was shorter among eyes co-managed with ATT, although statistically not significant ($p = 0.144$). Eyes experiencing at least one UME episode exhibited lower visual acuity at the last-follow-up than those without ($p = 0.020$).
Conclusions: Active vitreous cells, retinal vasculitis, and posterior uveitis are associated with UME among OTB patients. The time-to-resolution of UME for eyes co-managed with ATT was shorter compared to those without, suggesting that patients with UME in OTB should be treated with ATT.

Tubulointerstitial Nephritis with Uveitis syndrome in children: What to keep an eye on

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Introduction: Tubulointerstitial Nephritis with Uveitis (TINU) syndrome is an inflammatory disease involving both the eyes and kidney, most commonly seen in children. Due to its rarity, difficulties in diagnosis and prognosis still exist. The aim of this study is to describe the onset and clinical course to aid in recognizing and treating TINU.

Methods: We included a cohort of 32 children with TINU and retrospectively recorded clinical characteristics of both TIN (Tubulointerstitial Nephritis) and uveitis at first presentation and during a follow up of up to two years.

Results: Most children (84%) presented with ophthalmological symptoms. First presentation was often anterior uni-or bilateral uveitis, which in most patients progressed to chronic bilateral panuveitis. Patients that presented at the paediatrician first had more systemic symptoms and developed ocular symptoms up to 4 months after the onset of nephritis. There was a significant difference in time interval to inactivity under treatment in favour of nephritis compared to uveitis (p-value = 0.002). Twenty eight percent of patients developed chronic kidney disease (CKD) at 24 months with a suggestion of a beneficial effect of early treatment with corticosteroids.

Conclusions: Due to heterogeneity in presentation, screening for TINU is useful in all paediatric patients presenting with uni- and bilateral uveitis. Likewise, in patients with TIN, uveitis may occur after initial onset. Early diagnosis is important as uveitis requires long term treatment with immunosuppressive medication and TIN in turn may lead to CKD in these children.

Clinical patterns of sarcoidosis patients with and without uveitis: Insights from a Dutch sarcoidosis centre

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Purpose: Uveitis is a common ocular manifestation in individuals with sarcoidosis, a multisystem inflammatory disorder. This study aimed to explore clinical and genetic factors associated with the presence or absence of uveitis in sarcoidosis patients.

Methods: Total 625 Dutch sarcoidosis patients were included. 170 underwent ophthalmic examination, and 61 were diagnosed with uveitis. Demographic and clinical data, including age, gender, race, biopsy status, chest radiography findings, TNF-α inhibitor treatment, and uveitis classification were collected retrospectively from medical records. Genetic data was available for HLA haplotypes, *TNF-α* G-308A, and *BTNL2* G16071A polymorphisms.

Results: The majority of the patients presented with bilateral uveitis (80.3%). The proportion of women was higher in the uveitis group compared to the non-uveitis group (67.2% and 47.7%; $p = 0.014$). Pulmonary involvement (chest radiographic stage II-III) was significantly lower in patients with uveitis (36.1% versus 64.2%; $p < 0.001$). Patients with uveitis were more often treated with TNF-α inhibitors (67.2% versus 29.4%; $p < 0.001$) and the outcome was better compared with the non-uveitis group, 92% vs. 68%, responders ($p < 0.012$). Uveitis patients treated with TNF-α inhibitors (either adalimumab or infliximab) were more likely to suffer from intermediate or posterior uveitis than anterior uveitis. Genetic analysis identified a significant association between the *BTNL2* G16071A GG genotype and uveitis ($p = 0.012$).

Conclusion: This study highlights distinctive demographic, clinical and genetic features associated with uveitis in sarcoidosis patients. Ocular sarcoidosis was more prevalent in women. Further research is warranted to explore the implications for treatment strategies and prognostic assessments.

Novel polymer-vital dye combination for selective vitreous staining

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Introduction: Complete vitreous removal during vitrectomy is crucial to prevent recurrent retinal detachment. Adequate visualization of vitreous remnants at the vitreous base (VBR) and vitreoschisis-induced vitreous cortex remnants (VCR) on the retinal surface posterior to the vitreous base is essential, prompting research into improved staining methods. Various approaches have been explored, including triamcinolone acetonide crystals, lutein crystals, vital dyes, and fluorescein. Only triamcinolone acetonide is routinely used, as other methods lack selectivity or sufficient staining efficacy. This study aims to develop a more selective and effective method for vitreous staining.
Materials and Methods: Donor eyes were used to evaluate the staining properties of various polymer-dye combinations. Vitreous and retinal tissues were carefully dissected to ensure consistent assessment. Poly(2-ethyl-2-oxazoline) was tested with Trypan Blue and Chicago Sky Blue to determine its potential for selective staining. The polymer acts as a binding agent, forming a stable complex with the dyes, increasing their affinity for hyaluronic acid, a key vitreous component. Samples were exposed to the solutions under controlled conditions, and staining outcomes were visually evaluated for effectiveness and selectivity.
Results: Combining Trypan Blue or Chicago Sky Blue with poly(2-ethyl-2-oxazoline) effectively minimized non-specific staining of retinal structures. The polymer-dye complex selectively stained vitreous tissue in-depth, likely due to its interaction with hyaluronic acid.
Conclusion: Trypan Blue or Chicago Sky Blue combined with poly(2-ethyl-2-oxazoline) demonstrates potential as a selective and in-depth vitreous staining method. This approach could enhance visualization during vitrectomy, improving surgical outcomes.

Hyaluronan and chondroitin sulfate in the vitreous of patients with rhegmatogenous retinal detachment

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Purpose: To compare hyaluronan (HA) and chondroitin sulfate (CS) concentrations as well as the sulfation pattern of CS disaccharides in the vitreous of rhegmatogenous retinal detachment (RRD) versus non-RRD patients.
Methods: Vitreous samples from 56 patients with RRD and 41 patients with floaters, macular hole or epiretinal membranes, were collected during pars plana vitrectomy. Disaccharide analysis was performed by high-pressure liquid chromatography (HPLC) to assess HA and CS concentrations in the vitreous and determine CS sulfation. In addition, we used agarose gel-electrophoresis to evaluate the molecular mass of HA fragments.
Results: The vitreous of RRD patients showed significantly lower HA concentrations (Mann–Whitney U, $p<0.001$) and higher CS concentrations (Mann–Whitney U, $p<0.001$) compared to non-RRD patients. CS disaccharides were significantly more often non-sulfated in RRD patients, compared to control vitreous samples.
Conclusions: Decreased hyaluronan concentrations and relative more non-sulfated CS disaccharides in the vitreous may be related to RRD pathogenesis.

Efficacy and side effects of peroperative peribulbar betamethasone vs. triamcinolone during vitrectomy for epiretinal membrane peeling

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Purpose: Standard care was to administer subconjunctival betamethasone at the end of a vitrectomy for epiretinal membrane (ERM) to prevent cystoid macular edema (CME). From May 2023 triamcinolone was used instead, because betamethasone was not available. The aim of this study is to compare the efficacy and side effects between the two steroids.
Methods: Retrospectively, patients that underwent (phaco)vitrectomy for idiopathic ERM from October 2023 until October 2024 were included. Patients with a history of intraocular surgery other than cataract surgery, uveitis, or macular edema not related to an ERM

were excluded. The primary outcome was postoperative presence of CME and ocular hypertension.
Results: In total, 271 patients were included of which 140 patients received betamethasone and 131 patients received triamcinolone. All patients had at least 6 weeks follow-up. Baseline characteristics of the two groups, including grade of ERM, presence of intraretinal cysts, and visual acuity, were similar. Six weeks postoperatively, the central retina was significantly thicker in the betamethasone group (449 vs. 425 µm, $p<0.01$). In this group, significantly more patients had intraretinal cysts (64 (45%) vs. 43 (26%), $p<0.01$) and more patients were treated for CME (24 (17%) vs. 8 (6%), $p=0.05$) 6 weeks after surgery. Intraocular pressure (IOP), number of patients treated with IOP-lowering drugs, and visual acuity were not significantly different.
Conclusions: significantly more patients developed CME in the betamethasone group. There was no significant difference in IOP and visual acuity between the two groups.

Axial length and layer thickness in the Rotterdam Study using the EyeNED layer segmentation algorithm

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Purpose: This study aims to provide a comprehensive analysis of the relation between retinal layer thickness and AL in the general population using a device-agnostic AI algorithm.
Methods: An in-house AI model for layer segmentation was applied to all macular OCT scans in the Rotterdam Study, a prospective population-based cohort study. Eyes with visual impairment (visual acuity <0.5 decimal) were excluded. Automated fovea detection was applied and layer thickness measurements were obtained from the ETDRS grid, corrected using Littmann's method. Outliers (>3 standard deviations [SD]) and scans with severe decentration (>1 mm) were excluded. The relation between AL and layer thickness was assessed using multivariable linear regression, adjusting for age and sex. The analyses were also conducted in a subgroup of highly myopic eyes (AL >26 mm).
Results: We observed a consistent significant negative relationship between AL and thickness except for the retinal nerve fibre layer (RNFL). That was significantly thicker with increasing AL in the full cohort, but significantly thinner with increasing AL in the high-myopia subgroup.
Conclusions: Our results underscore the association between axial elongation and retinal thinning for most layers, except for RNFL. Retinal layer thickness should be interpreted in this context and be corrected using population-based benchmarking sets.

Phenotypic and genetic analysis of retinal vascular biomarkers in the rotterdam study

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Purpose: Fundus images allow for non-invasive assessment of the retinal vasculature; and AI makes it possible to analyse them at a large scale. This study provides a comprehensive analysis of 17 retinal vascular image-derived phenotypes (IDPs) in the Rotterdam Study (RS), including novel features like temporal angles, number of bifurcations, or diameter variability.
Methods: We extracted 17 IDPs from colour fundus images (CFIs) of 8142 subjects using an AI pipeline. We performed a Genome-Wide Association Study (GWAS) of these phenotypes and estimated heritability, phenotypic and genetic correlations and phenotype-disease correlations. We attempted replication of 566 SNPs previously discovered in the UK Biobank (UKBB) using the Benjamini-Hochberg procedure.
Results: IDP heritability ranged between 0 and 22%. Arterial tortuosity ($h^2=0.22\pm0.07$), ratio of arterial to venous tortuosity ($h^2=0.20\pm0.06$) and variability in venous diameter ($h^2=0.18\pm0.06$) had the highest heritability estimates. Cross-phenotype correlations mostly mirrored the corresponding phenotypic correlations. Phenotype-disease associations confirmed some previously known findings and revealed novel connections. Of 566 SNPs found in UKBB, 232 were replicated in the RS.
Conclusions: Large-scale, AI-based analysis of IDPs supported known genetic and phenotypic links and revealed new insights about the genetics of the retinal vasculature.

Population-based screening for glaucoma by means of a deep-learning algorithm on colour fundus images: A validation study

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Purpose: Glaucoma often remains undiagnosed in its early stages, with nearly half of all cases going undetected. Early detection and management are crucial to slow progression and prevent vision loss. This study validated a deep-learning (DL)-algorithm for classifying glaucoma on colour fundus images (CFIs) to explore the feasibility of large-scale screening.

Methods: The winning DL-algorithm from the prior AIROGS challenge was used. This DL-algorithm had been trained on 101 442 CFIs from 54 274 individuals across 486 sites in the USA with diverse ethnic representation. The DL-algorithm assessed the CFI gradability and identified referable glaucoma (RG), defined as glaucoma with presumed visual field loss, and no referable glaucoma (NRG). Validation utilized data from the Rotterdam Study (RS), a population-based cohort study in The Netherlands, comprising CFIs paired with visual field data. The criteria applied to the glaucomatous visual field loss (GVFL) classification align with those used in regular clinical settings. Model performance was evaluated by sensitivity at 95% specificity and the area under the receiver operating characteristic curve (AUC).

Results: Among RS participants aged 50–80, the DL-algorithm classified 5229 (3.9%) CFIs as ungradable. After selecting one follow-up moment, 18 777 CFIs were analysed, including 155 CFIs labelled with GVFL. The algorithm achieved a sensitivity of 74.2% at 95% specificity and an AUC of 0.921.

Conclusions: This study validates the DL-algorithm's adequate performance in detecting perimetric glaucoma in CFIs, highlighting the potential of DL-algorithms for population-wide glaucoma screening programs. High specificity ensures a reasonable positive predictive value, which is critical for population screening.

Deep learning-based quantification of intermediate age-related macular degeneration biomarkers for predicting geographic atrophy and neovascularization

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Purpose: The introduction of novel therapies onto the market, as well as those currently in the pipeline, has generated interest in large-scale population screening for age-related macular degeneration (AMD). We developed a deep learning (DL)-based algorithm for prediction modelling for conversion to geographic atrophy (GA) or neovascularization.

Methods: The prediction model analysis was based on 37 240 CFIs of 17 605 eyes collected from 8890 participants aged 45 and above in the Rotterdam Study, a prospective population-based cohort study. The model automatically quantified AMD biomarkers on CFIs and a random forest classifier was fitted to predict conversion to GA and/or neovascularization. The results were compared with human labels.

Results: A total of 313 eyes converted to late AMD, including 173 to GA, 172 to neovascularization, and 32 to mixed late AMD within a period of seven years. The DL-based prediction model achieved a similar AUC to human-graded labels in predicting conversion to GA (DL 0.867 (0.824–0.904) vs. human 0.893 (0.859–0.923)), while performing better in predicting neovascularization (0.821 (0.775–0.865) vs. human 0.730 (0.679–0.780)) and any late AMD (DL 0.883 (0.854–0.909) vs. human 0.849 (0.815–0.878)).

Conclusion: Predicting conversion from early or intermediate AMD to late AMD can be conducted highly accurately with the use of DL-based prediction modelling, achieving similar accuracy to human graders. Our algorithms can be used as a rapid and objective tool for prognostic modelling for individuals at risk, including screening, epidemiological studies, clinical management and patient selection for trials.

Dynamic prediction of late age-related macular degeneration using AI-based feature quantifications

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Purpose To dynamically predict late age-related macular degeneration (AMD) using artificial intelligence (AI)-based quantifications of intermediate AMD features over time.
Methods A total of 13.075 participants aged 45 years and older, from the prospective, population-based Rotterdam Study (RS) were included. Areas of drusen, hyperpigmentation, retinal pigment epithelium (RPE) degeneration, and reticular pseudodrusen on colour fundus photographs were measured at all follow-up visits with a deep learning algorithm developed by the EyeNED reading center. A subset of the data (n = 5.000 photographs) was used to develop a joint model for the dynamic prediction of the cumulative risk of late AMD.
Results A total of 325 participants developed late AMD. The mean follow-up duration was 8.5 years (standard deviation: 6.2 years). Areas of drusen, hyperpigmentation, RPE degeneration, and reticular pseudodrusen were associated with a higher cumulative risk of late AMD (estimates: 1.3 [95% confidence interval (CI): 0.3–1.8], 0.4 [95% CI: −1.7-2.0], 8.1 [95% CI: 5.6–10.3], and 0.7 [95% CI: 0.3–1.2], respectively). The joint model allowed for dynamic prediction of late AMD risk in RS participants who were not used in the training set (Figure 1).
Conclusion Repeated measures of AI-based quantified intermediate AMD features can be used to quickly, objectively, and dynamically predict progression to the late stages of the disease. Benefits of the dynamic prediction model include improved, up-to-date, and individualized prognoses for ophthalmologists and patients, as well as valuable insights for epidemiological research and treatment investigation in clinical trials.

Follow-up of cystic pineal glands in retinoblastoma patients does not increase detection of pineal trilateral retinoblastoma

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Purpose: To evaluate the effectiveness of baseline screening and follow-up with MRI to detecting trilateral retinoblastoma (TRb) and assessing the risk of TRb development.
Methods: Prospective multicenter cohort study, including 607 retinoblastoma patients (2012–2022) with followed up (9–2023). On MRI, we measured the height and width, evaluated the aspect (solid, partly cystic and completely cystic) of the pineal gland and evaluated radiological features suspicious of pineal TRb. The effectiveness of the current TRb screening method was assessed by evaluating its sensitivity and specificity to detect TRb.
Results: Heritable retinoblastoma patients had a risk of 3.78% to develop TRb. One out of four pineal TRbs was detected during a follow-up scan and four out of five non-pineal TRbs were detected on the baseline MRI. Screening for pineal TRb had a sensitivity of 25% and specificity of 100%, for non-pineal TRb the sensitivity was 80%. It required 494 follow-up scans to detect one pineal TRb. However, when restricting the follow-up to solely suspicious glands, only 22 scans were required to detect one pineal TRb.
Conclusion: At baseline MRI, only one pineal trilateral retinoblastoma was detected. Follow-up after 3 months should be restricted to patients with a suspicious pineal gland defined as irregularly thickening of the cyst wall (>2 mm), fine nodular aspect of the cyst wall or when a solid or cystic gland exceeds the upper 99% prediction interval for size.

The ever ongoing cosmetic quest to change eye colour

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Purpose: To provide a historical overview of the various efforts to change eye colour for purely cosmetic reasons, along with the associated potential risks and harms.
Methods: Literature and online search.
Results: Eye drops containing adrenaline were used during World War II on involuntary inmates of concentration camp Auschwitz-Birkenau. Prostaglandin drops, used in glaucoma therapy since 1996, can increase iris pigmentation as an undesired side-effect. Commercial drops, available since 2011, are not effective and potentially dangerous. Iris implants, commercially available since 2011, led to serious complications like corneal decompensation, uveitis and glaucoma, and for this reason, implants had to be removed, leaving some patients almost blind. Also commercially in use since 2011 are laser

treatments to make brown eyes blue. Among the possible complications are anterior uveitis and (pigmentary) glaucoma. Corneal tattooing has existed for almost 2000 years. Complications of modern, intrastromal keratopigmentation include corneal perforation, bacterial infection, allergic or toxic reaction to pigment, migration of pigment, and functional complications like visual field limitation and light sensitivity. Personal identity and self-esteem are likely contributing factors to undergo this potentially harmful cosmetic eye surgery. In addition to the earlier discussed complications, the artificial layer of colour can obscure ocular pathology of the cornea or iris. As the majority of individuals undergoing these procedures are relatively young, problems may arise in the future when they will need cataract or other eye surgeries.
Conclusion: Permanently changing eye colour in healthy eyes for purely cosmetic reasons is a risky procedure.